# Fast Multiplexed Superconducting-Qubit Readout with Intrinsic Purcell Filtering Using a Multiconductor Transmission Line

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Fast and accurate qubit measurement remains a critical challenge on the path to fault-tolerant quantum computing. In superconducting quantum circuits, fast qubit measurement has been achieved using a dispersively coupled resonator with a large externally limited linewidth. This necessitates the use of a Purcell filter that protects the qubit from relaxation through the readout channel. Here, we show that a readout resonator and filter resonator, coupled to each other both capacitively and inductively via a multiconductor transmission line, can produce a compact notch-filter circuit that effectively eliminates the Purcell decay channel through destructive interference. By utilizing linewidths as large as 42 MHz, we perform simultaneous readout of four qubits using a 56-ns integration window and benchmark an average assignment fidelity of 99.77%, with the highest qubit assignment fidelity exceeding 99.9%. Including the simulated readout ring-down time, the total readout duration was between 115 and 215 ns for the four qubits, which we anticipate can be reduced to around 100 ns with active ring-down pulse shaping. These results demonstrate a significant advancement in speed and fidelity for multiplexed superconducting-qubit readout.

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## I. INTRODUCTION

Quantum computing relies on the accurate measurement of qubit states. In superconducting quantum circuits, measurement now constitutes a major component of the error budget for algorithms that require detection and feedback. This measurement error is significant for the execution of quantum error-correction codes such as the surface code [1–6], which require the measurement of ancilla qubits at the end of every error-correction cycle. It is equally important in measurement-based protocols for preparing entangled states [7–11] and performing long-range entangling gates [12], where unitary qubit operations are replaced by midcircuit measurements and dynamical feedback.

Fast superconducting-qubit readout has been achieved by dispersively coupling each qubit to a readout resonator that possesses a large externally limited linewidth [13–17]. By utilizing a Purcell filter [18] to shield the qubit from environmental noise entering via the readout line, large resonator linewidths can be achieved without compromising the energy-relaxation time of the qubit. In a multiplexed readout circuit, a single Purcell filter can be shared by multiple readout resonators [13] or a dedicated Purcell filter can be assigned to each readout resonator. An example of the latter approach places a filter resonator between each readout resonator and the readout line [15], which provides Purcell filtering while simultaneously reducing measurement crosstalk.

However, as readout resonators with very large linewidths are explored [19] and as energy-relaxation times of superconducting qubits continue to improve well beyond 100 μs [20–26], the Purcell filtering in circuits employing filter resonators may be insufficient to ensure that the Purcell relaxation channel remains adequately suppressed. For example, in Ref. [19], the predicted Purcell-limited relaxation time with a dedicated filter resonator

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was 5 μs, which necessitated the introduction of an additional notch filter. This limited Purcell filtering motivates the design of filter resonators with an additional Purcell-filtering mechanism.

An attractive approach to improve Purcell filtering is to engineer an "intrinsic" filter that leverages properties of the circuit to enhance the filtering without the need for additional circuit components. Example intrinsic filtering mechanisms have involved exploiting symmetry in the qubit [27–29], making use of a stray capacitance in the readout circuit [30], and utilizing the distributed nature of the readout resonator [16] and the readout feedline [31].

Here, we propose and experimentally demonstrate a readout-resonator and dedicated-filter configuration featuring an intrinsic notch-type filter. The notch filtering is achieved by coupling two quarter-wavelength ($\lambda/4$) coplanar waveguide (CPW) resonators together through a multiconductor transmission line (MTL) [32], which is formed by bringing a segment of the resonator lines into close proximity. The capacitive and inductive interactions in the MTL interfere destructively, resulting in a notch frequency at which signals cannot propagate through the resonators. We experimentally demonstrate that this filtering mechanism can effectively eliminate the Purcell relaxation channel when the notch is tuned to the qubit frequency. By utilizing readout modes with large effective linewidths, we then perform frequency-multiplexed readout of four qubits using a sub-60-ns measurement pulse and integration window, with the highest qubit assignment fidelity exceeding 99.9%.

The paper is organized as follows. First, a distributed-circuit model and an equivalent lumped-element circuit model for the coupled resonators are presented and utilized to find analytic expressions for the notch frequency and the coupling strength between the resonators. Next, the notch filtering is experimentally demonstrated and the predictions of the analytic formulas are verified. Finally, fast four-to-one multiplexed readout is characterized in a device featuring readout and filter resonators with intrinsic notch filtering.

## II. MODEL

A coupled-mode representation of dispersive readout with a dedicated filter resonator is shown in Fig. 1(a). A qubit is dispersively coupled to a readout resonator with coupling strength $g$, which in turn is coupled to a filter resonator with coupling strength $J$. The filter resonator is then coupled with an externally limited linewidth of $\kappa_p$ to a readout line. It is generally advantageous to minimize the detuning between the readout resonator and filter resonator in order to increase the effective linewidth of the readout resonator [17,33].

In Fig. 1(b), we introduce the readout- and filter-resonator configuration used here to realize an intrinsic notch filter in the readout- and filter-resonator circuit. Two $\lambda/4$ resonators formed from CPWs with characteristic impedance $Z_0$ and phase velocity $v$ are coupled together through a length-$\ell_c$ section where the lines are brought into close proximity. The total lengths of the readout and filter resonators are $\ell_r = \ell_r^o + \ell_c + \ell_r^s$ and $\ell_p = \ell_p^o + \ell_c + \ell_p^s$, respectively, and their fundamental $\lambda/4$ resonances are at $\omega_{r(p)} = \pi v/(2\ell_{r(p)})$. The superscript "$o$" ("$s$") refers to the open (shorted) part of the lines.

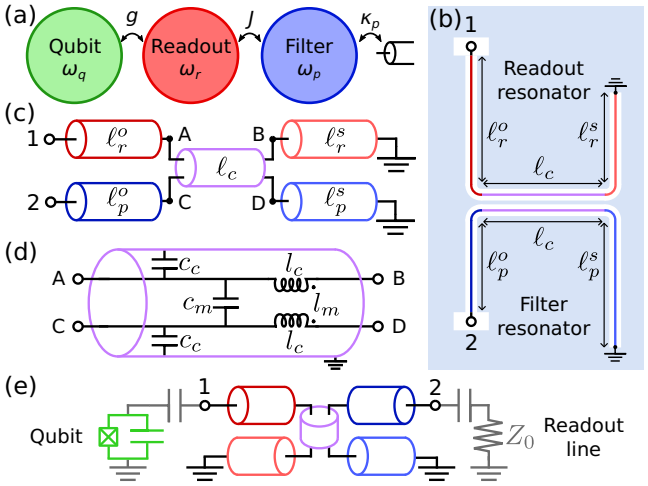



FIG. 1. Readout resonator and filter resonator with additional notch filtering. (a) Coupled-mode representation of dispersive readout with a readout resonator and dedicated filter. (b) CPW configuration used in this work to construct a readout resonator and filter resonator with additional notch filtering. (c) Distributed-circuit-model representation. The coupled section is treated as an MTL with two coupled lines. (d) Model of the MTL with the per-length parameters defined. (e) Depiction of a qubit coupled to the readout line through the distributed readout- and filter-resonator circuits.

The coupled-mode representation of Fig. 1(a) is valid for analysis near the readout- and filter-mode frequencies. However, it is known to be inaccurate for analyzing the Purcell filtering of the qubit, as this depends on the response of the readout circuit at the qubit frequency [34]. To analyze the Purcell filtering, we use a distributed-circuit-model representation, which provides a more accurate depiction of the readout-line environment as seen by the qubit.

In Fig. 1(c), we present the distributed circuit used to model the coupled resonators. The length-$\ell_c$ region where the coupling occurs is treated as an MTL, with parameters defined in Fig. 1(d). The two lines of the MTL are coupled together by the capacitance and inductance per length, $c_m$ and $l_m$, respectively. Both lines are symmetric with the same capacitance to ground and self-inductance per length, $c_c$ and $l_c$, respectively.

By making appropriate assumptions about the MTL section, an analytic expression for the transfer impedance

$Z_{21}(\omega) \equiv V_2(\omega)/I_1(\omega)|_{I_2=0}$ between the two ports at the open ends of both resonators is derived [see Eq. (B7) in Appendix B]. At frequencies at which $Z_{21} = 0$, the qubit in Fig. 1(e) is decoupled from the readout line because no amount of excitation applied at port 1 of the readout resonator can induce a voltage at port 2 of the filter resonator. Using this analytic expression, we analyze the behavior of the two resonators in the cases in which they are capacitively and MTL coupled. The capacitive-coupling case serves as a reference for quantifying the improvement in Purcell filtering achieved with MTL coupling.

### A. Capacitive coupling

Capacitive interaction between two $\lambda/4$ CPW resonators, i.e., a readout resonator and filter resonator, as depicted in Fig. 2(a), has been widely used for superconducting-qubit readout [4,5,15,17,35]. A lumped-element capacitor represents a special case of the MTL depicted in Fig. 1(d), corresponding to the limits $c_m \ell_c \to C_J$ and $l_m, \ell_c \to 0$. In this case, the solution to $Z_{21}(\omega)$ is given by (for details, see Appendix B)

$$Z_{21}(\omega) = -iZ_0^2 \frac{\sin\left(\frac{\omega \ell_r^s}{v}\right) \sin\left(\frac{\omega \ell_p^s}{v}\right)}{\cos\left(\frac{\pi\omega}{2\omega_r}\right) \cos\left(\frac{\pi\omega}{2\omega_p}\right)} \omega C_J. \tag{1}$$

The lowest nonzero frequency at which $Z_{21}(\omega) = 0$ is greater than or equal to $\min(2\omega_r, 2\omega_p)$, confirming that it is not possible to engineer a notch filter in the relevant frequency range for Purcell filtering by using a purely capacitive coupling.

### B. MTL coupling

MTL coupling has been previously employed in superconducting quantum circuits to couple CPW resonators to other CPW resonators [13,36] and to a feedline [31, 37]. However, as far as we are aware, it has always been approximated as a lumped-capacitive or a lumped-inductive coupling, depending on the circuit configuration,

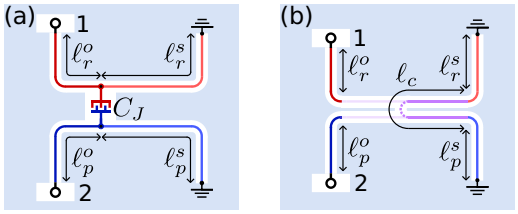



FIG. 2. Schematics of coupled $\lambda/4$ CPW resonators. (a) Diagram of lumped-element capacitive coupling between a pair of $\lambda/4$ resonators. (b) Geometric interpretation of the intrinsic notch that arises between a pair of MTL-coupled $\lambda/4$ resonators. The notch occurs at the antiresonance frequency of the effective $\lambda/2$ resonator formed between the two resonators, having a total length of $\ell_r^s + \ell_c + \ell_p^s$.

whereas here we fully account for the distributed capacitive and inductive coupling. We make use of the fact that CPWs on a substrate with permittivity $\epsilon_r$ behave to a good approximation as if they were embedded in a homogeneous medium with an effective permittivity of $\epsilon_{\text{eff}} = (1 + \epsilon_r)/2$ [38]. By applying this result to the MTL that couples the resonators, the solution to $Z_{21}(\omega)$ becomes (for details, see Appendix B)

$$Z_{21}(\omega) = iZ_0 \frac{\sin\left(\frac{\omega \ell_c}{v}\right) \cos\left(\frac{\pi\omega}{2\omega_n}\right)}{\cos\left(\frac{\pi\omega}{2\omega_p}\right) \cos\left(\frac{\pi\omega}{2\omega_r}\right)} \left(\frac{c_m}{c}\right), \tag{2}$$

where $c = 1/(Z_0 v)$ is the capacitance to ground per length of the CPW lines. The explicit dependence on the coupling inductance per length $l_m$ has dropped out because in a homogeneous medium, it is fully determined by the other line properties [32]. Here, it satisfies $l_m = Z_0^2 c_m$. The lowest nonzero frequency at which $Z_{21}(\omega) = 0$, hereafter referred to as the "notch frequency" $\omega_n$, takes the form

$$\omega_n = \frac{\pi v}{2\left(\ell_r^s + \ell_c + \ell_p^s\right)}. \tag{3}$$

The notch frequency matches the antiresonance of the effective $\lambda/2$ resonator with length $\ell_r^s + \ell_c + \ell_p^s$ that is formed between the shorted ends of the readout and filter resonators, as depicted in Fig. 2(b). Due to its independence from the lengths $\ell_r^o$ and $\ell_p^o$, the notch frequency can be adjusted independently from the readout- and filter-resonator frequencies and can be designed at both positive and negative detunings from these modes. As a result, it can be tuned for use as a notch filter to suppress the Purcell relaxation channel for a qubit in both positive- and negative-detuning regimes.

In order to simplify the analysis of the coupled resonators and to determine an expression for the coupling strength between them, we find an approximate mapping from the distributed-circuit representations in Fig. 3(a) to the lumped-element representations in Fig. 3(b). These lumped-element circuits accurately reproduce the electrical responses of the distributed circuits at frequencies around the $\lambda/4$ modes. The equations governing the mapping are provided in Appendix C. The $\lambda/4$ modes are converted to parallel $LC$ resonators, the lumped-element capacitance $C_J$ in Fig. 3(a) (i) is transformed to the capacitance $\tilde{C}_J$, and the MTL coupler in Fig. 3(a) (ii) is transformed to a parallel $LC$ resonator with capacitance $C_n$ and inductance $L_n$, satisfying $1/\sqrt{L_n C_n} = \omega_n$ such that the resonator frequency equals the notch frequency. The destructive interference between $C_n$ and $L_n$ ensures that the transfer impedance goes to zero at the notch frequency, reproducing the behavior of the distributed circuit.

The $Z_{21}(\omega)$ values for the equivalent lumped-element circuits and the exact solution to the distributed circuits

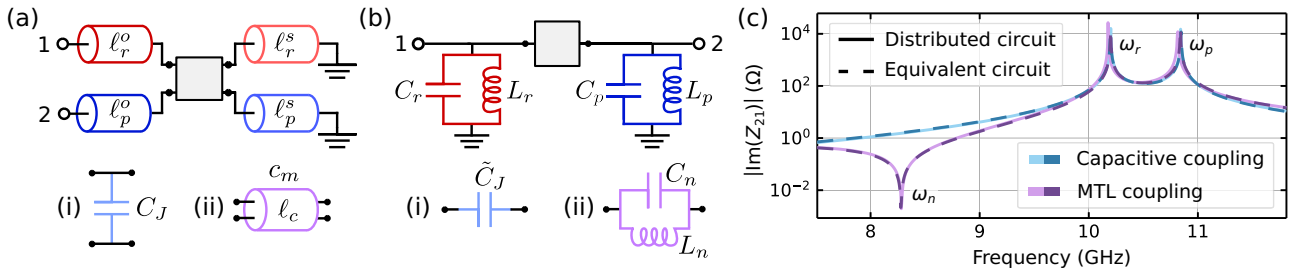



FIG. 3. (a) Distributed-circuit model for the coupled $\lambda/4$ resonators. The coupling element (gray square) represents (i) a lumped capacitor or (ii) an MTL. (b) Equivalent lumped-element circuit models. The coupling element is represented as (i) a lumped capacitor in the case of direct capacitive coupling or (ii) an $LC$ resonator in the case of MTL coupling. (c) Imaginary part of the transfer impedance $Z_{21}$ of the capacitive and MTL coupling circuits for the set of parameters given in Table III in Appendix C. The solid curves are the frequency dependence obtained from the distributed-circuit models and the dashed curves are from the equivalent-circuit models. The coupling strength between the resonators is $J/2\pi = 30$ MHz in both cases.

are compared in Fig. 3(c) for a given set of parameters, showing excellent correspondence at frequencies around the notch frequency $\omega_n$ and the $\lambda/4$ modes. We infer that the low-energy Hamiltonian for the distributed circuit closely matches the equivalent-circuit Hamiltonian, given the close correspondence of their electrical responses. The interaction Hamiltonian for the equivalent circuit with the MTL coupler, depicted in Fig. 3(b) (ii), takes the form (for the derivation, see Appendix C)

$$\hat{\mathcal{H}}_{\text{int}} = \hbar J(\hat{a}_r^\dagger \hat{a}_p + \hat{a}_r \hat{a}_p^\dagger), \tag{4}$$

$$J = \overline{\omega}_{rp} \frac{\pi^2}{32} \frac{\left(\frac{\overline{\omega}_{rp}}{\omega_n} - \frac{\omega_n}{\overline{\omega}_{rp}}\right)^3}{\cos^2\left(\frac{\pi\omega_n}{2\overline{\omega}_{rp}}\right)} \left(\frac{c_m}{c}\right) \sin\left(\frac{\omega_n \ell_c}{v}\right), \tag{5}$$

where $\overline{\omega}_{rp} = (\omega_r + \omega_p)/2$ is the average of the readout- and filter-resonator frequencies and we have taken $|\omega_r - \omega_p| \ll \omega_n$. The terms $\hat{a}_{r(p)}^\dagger$ and $\hat{a}_{r(p)}$ are the creation and annihilation operators for the readout- and filter-resonator modes and we have used the rotating-wave approximation (RWA) to drop the non-photon-number-preserving terms. The exchange coupling $J$ is suppressed as the notch frequency approaches $\overline{\omega}_{rp}$, scales linearly with the capacitance ratio $c_m/c$, and similarly scales linearly with the coupled-section length $\ell_c$, given that $\omega_n \ell_c \ll v$. The capacitance ratio $c_m/c$ can be determined efficiently using a conformal-mapping technique [38], which allows $J$ to be predicted quickly for a given set of line dimensions. The dependence of $c_m/c$ on the width $d$ of the ground strip separating the coupled lines is plotted in Fig. 4(c). The notch frequency is, to a good approximation, independent of $d$, per Eq. (3). Thus, this width provides a parameter to tune the $J$ coupling independently of the notch frequency.

### C. Purcell-filtering enhancement

In order to quantify the effect of the notch on the Purcell filtering, we compare the predicted Purcell-limited energy-relaxation times for a qubit that is coupled to the readout line through a readout resonator and filter resonator that are MTL coupled and capacitively coupled. The coupling strength $J$ is assumed to be the same. We define the Purcell-filtering enhancement factor $\xi$ as the ratio of these relaxation times. Employing the equivalent circuits from the previous section, the enhancement factor is given by (for the derivation, see Appendix D)

$$\xi = \frac{1}{4}\left(\frac{\omega_q}{\Delta_{qn}}\right)^2 \left(1 - \frac{\omega_n^2}{\overline{\omega}_{rp}^2}\right)^2, \tag{6}$$

where $\omega_q$ and $\Delta_{qn} = \omega_q - \omega_n$ ($\Delta_{qn} \ll \omega_n$) are the qubit frequency and the detuning between the qubit and the notch, respectively. The enhancement diverges like $1/\Delta_{qn}^2$ as the qubit approaches the notch frequency. The expression can also be rearranged into a formula for the bandwidth $B$ around the notch, where the enhancement to the Purcell-limited relaxation time of the qubit is at least a factor of $\xi$,

$$B \approx \frac{\omega_n}{\sqrt{\xi}} \left|1 - \frac{\omega_n^2}{\overline{\omega}_{rp}^2}\right|. \tag{7}$$

This expression predicts that for the device parameters in the next section, the Purcell filtering is enhanced by at least 2 orders of magnitude over bandwidths exceeding 200 MHz around the notch frequency. In Appendix D, an expression for the total Purcell-filtering enhancement factor including the contribution of the filter resonator is provided.

## III. EXPERIMENTAL RESULTS

### A. Purcell protection of the qubit relaxation

We have fabricated a 16-qubit device with each qubit coupled to a readout resonator and dedicated filter resonator, featuring MTL coupling and notch frequencies that

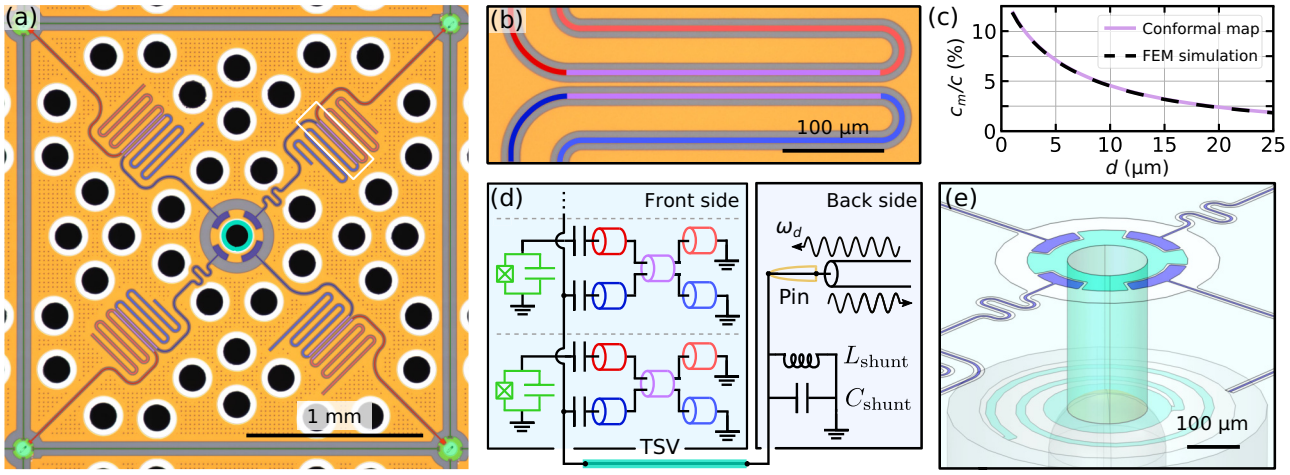



FIG. 4. (a) False-colored image of a four-qubit unit cell from a 16-qubit device. The four transmon qubits (green) are coupled to their nearest neighbors by capacitive buses (dark green). The four $\lambda/4$ readout resonators (red) couple to $\lambda/4$ dedicated filter resonators (blue), which couple to a shared readout line on the circuit back side through a TSV (teal). (b) Enlargement of the region enclosed by the rectangle in (a), showing the MTL section (purple) that couples the readout resonator and filter resonator. The width of the ground strip separating the two center lines is $d = 3.8\,\mu$m. (c) Ratio of the per-length capacitances $c_m$ and $c$ as a function of the ground strip width $d$, determined numerically using a conformal-mapping technique (solid) and from finite-element (FEM) simulation using COMSOL [39] (dashed). (d) Circuit diagram of the readout configuration for the unit cell. (e) Schematic three-dimensional transparent image of the TSV structure (teal) that routes signals from a pogo pin (yellow) on the circuit back side to the filter resonators on the circuit front side. The TSV structure features a spiral inductor patterned on the circuit back side, which screens the parasitic shunt capacitance of the TSV structure.

are tuned to the respective target qubit frequencies. The device architecture is based on a four-qubit unit cell that is tiled to create an array of fixed-frequency transmon qubits with nearest-neighbor couplings. In Fig. 4(a), we show one of the four-qubit unit cells, featuring TiN-based transmon qubits and resonators, and superconducting through-silicon vias (TSVs) metalized with Al. The TSVs connect the front-side ground planes via a common back-side ground plane and as a result air bridges [40] are not used. Readout is performed by a reflection measurement through a pogo pin that contacts a TSV on the back side of the chip. Signals are routed to the filter resonators through the TSV as depicted in Fig. 4(d). In Fig. 4(e), we show a scale diagram of the through-substrate readout design. The TSV structure has a large (approximately 230 fF) parasitic shunt capacitance to ground, which reduces the linewidth of the filter resonators. In order to mitigate this, a spiral shunt inductor is patterned on the circuit back side to screen the shunt capacitance around the readout frequencies (for details, see Appendix E).

In Fig. 5(a), we show the reflected phase response of a signal applied to the readout pogo pin on the circuit back side. Due to the relatively small detuning between the filter resonators compared to their linewidths, the parameters of the four readout resonators and four filter resonators are fit simultaneously to a reflection model that encompasses all eight resonators in the four-qubit unit cell (for details, see Appendix E). By fitting the $|g\rangle$- and $|e\rangle$-state responses simultaneously, the dispersive shifts of the readout resonators are also determined. The readout-mode parameters determined from the fit, together with the qubit parameters for qubits $Q_1$–$Q_4$ determined from their spectroscopy, are presented in Table I. Here, the readout modes refer to the hybridized modes formed between the readout and filter resonators, which have been addressed during the qubit readout (for details, see Appendix E). The bare readout- and filter-resonator parameters are also presented in Table IV in Appendix E. We note that the target value for the linewidth of the readout modes was $\kappa_{\mathrm{ro}}^{g}/2\pi = 40\,\mathrm{MHz}$ for all qubits and that the measured variation is due to the sensitivity of this parameter to the detuning between the readout and filter resonators.

The Purcell protection of each qubit is characterized following the method in Ref. [16]. First, the drive power $P$ incident on the device is calibrated by measuring the ac Stark shift of the qubit caused by a readout tone (see Appendix F for details). The drive amplitude $\Omega$ incident on the qubit is then determined by two methods: (i) by measuring the induced Rabi-oscillation frequency in the qubit when the drive was on resonance with the $|g\rangle$-$|e\rangle$ and $|e\rangle$-$|f\rangle$ transitions and (ii) by measuring the drive-induced Stark shift on the qubit when the drive was detuned from these transitions. The Purcell-limited relaxation time $T_1^{\mathrm{pl}}$ over a range of frequencies is then determined from the ratio of $P$ and $\Omega^2$ (for details, see Appendix F). The Purcell-limited relaxation times for the

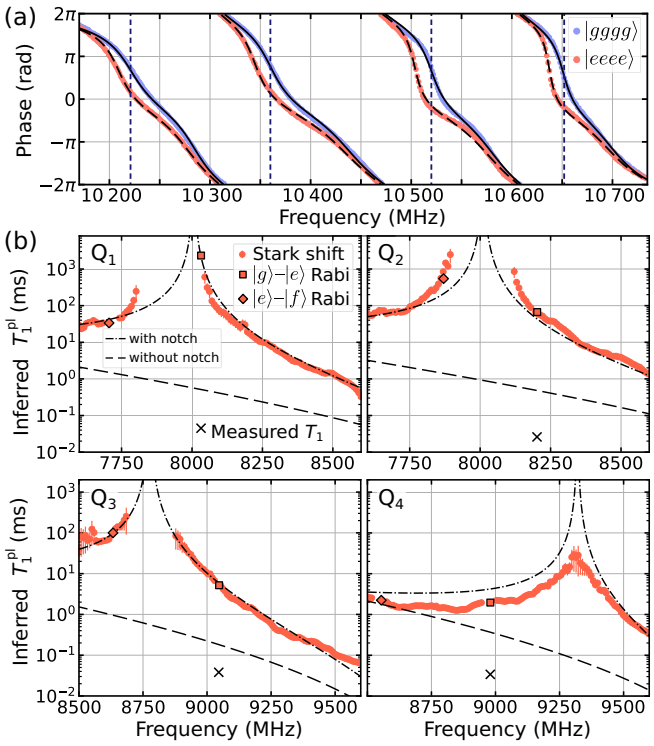



FIG. 5. (a) Reflection spectra of the four readout-resonator and filter-resonator pairs for the four qubits prepared in the $|g\rangle$ (blue) and $|e\rangle$ (red) states. The phase is wrapped modulo $4\pi$ to better visualize the response of each readout- and filter-resonator pair. The solid and dashed lines show the fitted phase response for the $|g\rangle$- and $|e\rangle$-state preparations, respectively. The vertical dashed lines show the four readout-mode frequencies $\omega_{\rm ro}^g$. (b) Purcell-limited relaxation time $T_1^{\rm pl}$ of the four qubits as a function of frequency inferred from the Rabi oscillations driven at the qubit $|g\rangle$-$|e\rangle$ and $|e\rangle$-$|f\rangle$ transition frequencies (squares and diamonds, respectively) and the ac Stark shift induced by the drive detuned from these transitions (dots). The dot-dashed and dashed black lines show the prediction of a circuit model with and without the intrinsic notch filtering, respectively. The black crosses show the measured qubit relaxation time.

four qubits are shown in Fig. 5(b). The predicted values using an equivalent-circuit model of the readout with and without the intrinsic notch filtering are overlaid (for details, see Appendix D). The notch frequency is the only fitting parameter in these models, with the circuit otherwise determined from the qubit and readout parameters. The Purcell-limited relaxation times at the $|g\rangle$-$|e\rangle$ transition frequency and the predicted relaxation enhancement factors $\xi$ are presented in Table I. The evaluated Purcell-limited relaxation times all exceed 2 ms and are predicted to be enhanced by at least an order of magnitude due to the intrinsic notch filtering. The qubit-notch detunings are $\Delta_{\rm qn}/2\pi \approx \{30, 180, 270, -350\}$ MHz for qubits $Q_1$–$Q_4$. In the case of qubit $Q_1$, the notch frequency is nearly on resonance with the qubit frequency and as a result the

TABLE I. Qubit and readout parameters. Here, $\omega_q$ and $\alpha$ are the qubit frequency and anharmonicity, respectively. The terms $\omega_{\rm ro}^g$ and $\omega_{\rm ro}^e$ are the hybridized readout-mode frequencies given that the qubit is in the $|g\rangle$ and $|e\rangle$ state, respectively, and $\chi_{\rm ro}$ is the readout-mode dispersive shift defined as $\omega_{\rm ro}^e \equiv \omega_{\rm ro}^g + 2\chi_{\rm ro}$. The term $T_1^{\rm pl}$ is the Purcell-limited relaxation time evaluated at the qubit frequency and $\xi$ is the predicted Purcell-filtering enhancement factor due to the intrinsic notch filter.

| | $\omega_q/2\pi$ (MHz) | $\alpha/2\pi$ (MHz) | $\omega_{\rm ro}^g/2\pi$ (MHz) | $\kappa_{\rm ro}^g/2\pi$ (MHz) | $\chi_{\rm ro}/2\pi$ (MHz) | $T_1^{\rm pl}$ (ms) | $\xi$ |
|---|---|---|---|---|---|---|---|
| $Q_1$ | 8032 | −326 | 10 221 | 42 | −5.9 | 2300 | ∼ 3000 |
| $Q_2$ | 8189 | −333 | 10 360 | 34 | −7.8 | 70 | 85 |
| $Q_3$ | 9046 | −414 | 10 520 | 24 | −8.3 | 5 | 25 |
| $Q_4$ | 8980 | −425 | 10 652 | 19 | −7.1 | 2 | 10 |

inferred value of $T_1^{\rm pl}$ is 2.3 s even though the effective readout linewidth is 42 MHz, demonstrating the effective elimination of the Purcell relaxation channel. For qubits $Q_2$ and $Q_3$, the larger qubit-notch detunings are caused by deviation of the notch frequencies from their target values, whereas in the case of $Q_4$ it is due to a deviation of the qubit frequency from its target value.

In Fig. 6, we compare the measured notch frequencies and coupling strengths $J$ to the predictions of Eqs. (3) and (5), respectively. The phase velocity $v$ and the capacitance ratio $c_m/c$ in the analytic expressions are determined numerically by conformal mapping. The measured values of the notch frequencies $\omega_n$ are used in Eq. (5) to calculate the $J$ values, in order to assess the accuracy of the two expressions independently. The measured notch frequencies and $J$ couplings on average differ from the predicted values by 5% and 19%, respectively. The underestimation of the coupling strengths is in part due to the assumption that coupling only occurs along the MTL-section-length $\ell_c$ as defined in Fig. 4(b), since this neglects the coupling between the curved sections of the lines.

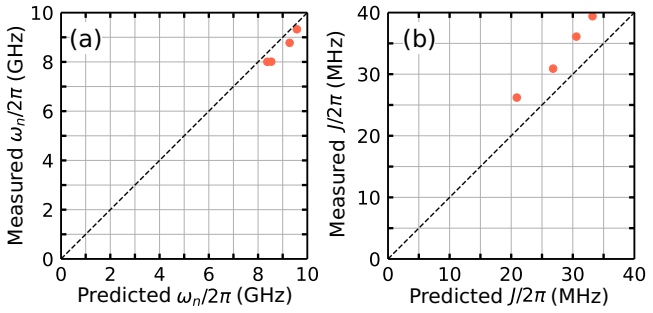



FIG. 6. Evaluation of the analytic solutions for (a) the predicted notch frequency $\omega_n$ and (b) the coupling strength $J$. The measured values are compared to the predictions of Eqs. (3) and (5) for the four readout- and filter-resonator pairs.

### B. Multiplexed fast readout

In Fig. 7(a), we show the time dependence of the separation $S(t) \equiv |s^e_{\rm out}(t) - s^g_{\rm out}(t)|$ of the qubit-state-dependent coherent output fields $s^{g(e)}_{\rm out}$ from the device in response to a 100-ns-long rectangular tone applied at the optimal readout frequency of qubit $Q_2$. The measured state-dependent heterodyne signals have been converted to the separation $S(t)$ through the relationship $\Gamma_m = S^2_{\rm ss}/2$ between the measurement-induced dephasing rate $\Gamma_m$ and the steady-state separation $S_{\rm ss}$ [41]. The output field separation is related to the signal-to-noise ratio (SNR) of the measurement through the expression

$$\mathrm{SNR} = \sqrt{2\eta \int_0^{\tau_{\rm meas}} S(t)^2\, dt}, \tag{8}$$

where $\eta$ is the quantum efficiency of the readout chain and $\tau_{\rm meas}$ is the duration of the integration window. This expression assumes that optimal weight functions have been applied to the integrated signal [42]. The measurement displays a short transient period of around 50 ns before the output field carries the maximum information about the qubit state. A semiclassical model for the output-field separation (see Appendix E) evaluated using the independently determined readout- and filter-resonator parameters shows good correspondence to the measured dynamics.

The rapid rise in output-field separation enables the tune-up of fast multiplexed readout on the four qubits with high fidelity. For single-shot readout, a flux-driven impedance-matched parametric amplifier [43] is used in the phase-preserving mode of operation in order to enable simultaneous amplification of measurement signals targeting the four readout resonators. At the optimal point, the quantum efficiency $\eta$ is determined to be 27%, 46%, 40%, and 42% for qubits $Q_1$–$Q_4$, respectively, by comparing the measurement-induced dephasing to the measured SNR [42].

In Figs. 7(b) and 7(c), we show the pulse sequences used to benchmark the readout assignment fidelity $F$ and the quantum nondemolition (QND) fidelity $F_Q$, respectively. These sequences check for the $|g\rangle \to |e\rangle$ and $|e\rangle \to |g\rangle$ error processes but will fail to detect measurement-induced leakage errors [44–46] when leakage states are misclassified as the target state by the two-state classifier. To reliably detect leakage errors, further pulse sequences such as pseudosyndrome detection could be applied [29]. The pulse sequences are carried out on the four qubits simultaneously and repeated $4 \times 10^4$ times both with and without the $\pi$ pulse. The two-step measurement pulses feature an initial 14-ns flat top with a 6-ns raised-cosine edge and a flat-top amplitude between 1.35 and 1.4 times higher than that of the subsequent flat plateau in order to increase the output-field separation at early times. The steady-state photon-number populations of the readout resonators during the main pulse step, $n^g_r$ (given that the qubit is in the $|g\rangle$ state), are provided in Table II as a fraction of the critical photon number [47]. The acquired measurement signals are integrated over an optimally weighted time window of 56 ns. The qubit states are assigned from the integrated signals using a two-state logistic regression discrimination model [48] that has been trained on a prior data set of $2 \times 10^4$ single-shot measurements. The assignment fidelities are determined using the definition

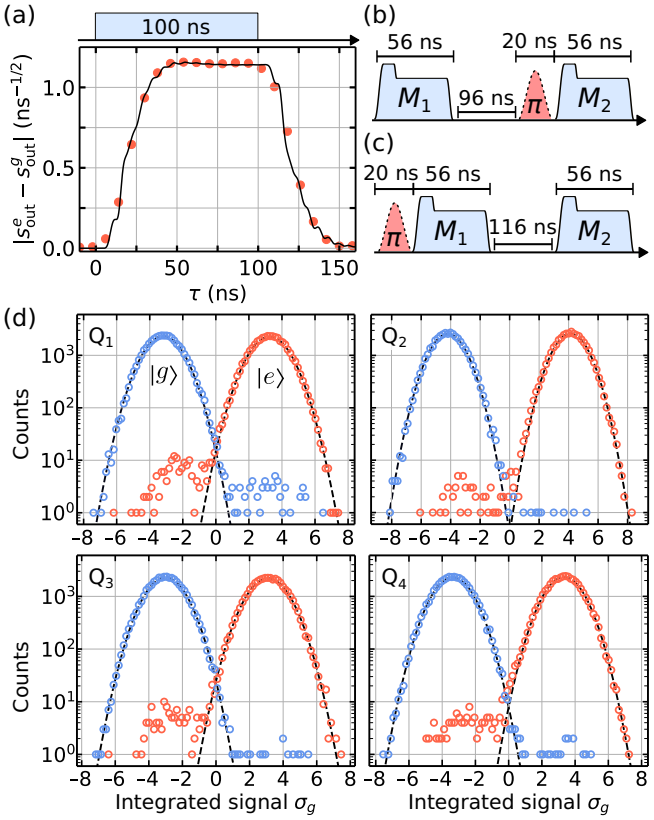



FIG. 7. Demonstration of fast readout. (a) Time dependence of the output-field separation $S(t) \equiv |s^e_{\rm out}(t) - s^g_{\rm out}(t)|$ in response to a 100-ns-long rectangular readout pulse at the optimal readout frequency of qubit $Q_2$, averaged over $4 \times 10^4$ repetitions. The solid black line is the prediction of the semiclassical model for the coherent output fields. (b),(c) Readout-pulse sequences used to determine the assignment and QND fidelities, respectively. The readout sequences are performed with and without the 20-ns-long $\pi$ pulse. (d) Histograms of simultaneous time-integrated single-shot signals for the four qubits projected along the principal axes in the in-phase and quadrature (IQ) plane. The dashed lines show the fits of the $|g\rangle$- and $|e\rangle$-state signal distributions to Gaussian distributions. The horizontal axes are normalized by the standard deviation $\sigma_g$ of the $|g\rangle$-state Gaussian distribution.

TABLE II. Readout results. Here, $n_r^g/n_{\mathrm{crit}}$ represents the ratio of the steady-state photon number in the readout resonator during the readout pulse to the critical photon number. The terms $\varepsilon$ and $\varepsilon_Q$ are the assignment and QND errors, respectively, defined as $\varepsilon \equiv 1 - F$ and $\varepsilon_Q \equiv 1 - F_Q$. The terms $\varepsilon_{\mathrm{cl}}$ and $\varepsilon_{\mathrm{cl}}^Q$ are the coherence-limited error budgets on the assignment and QND fidelity, estimated from measurements of the qubit relaxation time carried out within an hour of the readout benchmarking. Details of the coherence limits and separation error calculations are given in Appendix G.

| | $n_r^g/n_{\mathrm{crit}}$ | $\varepsilon$ (%) | $\varepsilon_{\mathrm{cl}}$ (%) | $\varepsilon_Q$ (%) | $\varepsilon_{\mathrm{cl}}^Q$ (%) | $\varepsilon_{\mathrm{sep}}$ (%) |
|---|---|---|---|---|---|---|
| $Q_1$ | 1.50 | 0.35(2) | 0.03 | 1.91(5) | 0.19 | 0.08 |
| $Q_2$ | 1.05 | 0.09(1) | 0.05 | 0.70(3) | 0.33 | <0.01 |
| $Q_3$ | 0.82 | 0.29(2) | 0.04 | 1.10(4) | 0.23 | 0.13 |
| $Q_4$ | 0.89 | 0.20(2) | 0.04 | 0.55(3) | 0.25 | 0.04 |

$F \equiv [P_0(g_2 \mid g_1) + P_\pi(e_2 \mid g_1)]/2$, where $P(a \mid b)$ denotes the probability of event $a$ given event $b$. The subscripts 1 and 2 refer to the first and second measurement outcomes and the subscripts $\pi$ and 0 refer to the pulse sequences in Fig. 7(b) with and without the $\pi$ pulse. The QND fidelities are determined from the QND pulse sequences in Fig. 7(c) using the definition $F_Q \equiv [P_0(g_2 \mid g_1) + P_\pi(e_2 \mid e_1)]/2$. The resulting assignment and QND errors are summarized in Table II, along with the predicted coherence-limited error imposed by qubit relaxation.

In Fig. 7(d), we show the histograms formed by projecting the single-shot IQ points from the second measurement of the assignment fidelity pulse sequence along the principal axis. The results are postselected to ensure that the qubit was in the $|g\rangle$ state prior to the $\pi$ pulse. The histogrammed signals closely match Gaussian distributions, with outliers that are discussed below. From the overlap of the Gaussian fits, the state-separation errors $\varepsilon_{\mathrm{sep}}$ are calculated and summarized in Table II.

The average assignment fidelity for the simultaneous measurement is 99.77%, representing a significant advancement for fast multiplexed qubit readout [15]. Focusing on qubit $Q_2$, the separation error is negligible and the assignment fidelity is 99.91(1)%. The probability of measurement-induced excitation from $|g\rangle$ to $|e\rangle$ is low, with $P_0(e_2 \mid g_1) = 0.03\%$. In Appendix G, the IQ plot is examined, showing that less than 0.01% of measurements exhibit discernible excitation to leakage states. The assignment and QND fidelity are 0.04% and 0.37% below their coherence limits, respectively, implying some degree of measurement-induced relaxation [49,50]. On the other hand, the assignment and QND fidelities for qubit $Q_1$ (which had the lowest quantum efficiency) and $Q_3$ are markedly lower than their respective coherence limits, implying that the readout pulse causes significant measurement-induced excitation and relaxation. The IQ plot for $Q_1$ reveals measurement-induced excitation to leakage states that contributes significantly to the assignment error (see Appendix G). This leakage could be countered with improved parametric amplification at the readout frequency of $Q_1$, which would facilitate readout at a reduced photon number.

After application of the 56-ns measurement pulse, a simulation of the output field predicts that between 60 ns and 160 ns elapses before the readout-resonator photon numbers $n_r^{g,e}$ decrease to below $10^{-4}$, where the dephasing effect on future qubit operations is negligible, setting a total readout time of approximately 115–215 ns. From simulation, we anticipate that this total readout time can be reduced to around 100 ns by actively depopulating the readout resonators after the measurement pulse [19,29,51]. Further increasing the assignment fidelities beyond 99.9% will require increasing the qubit relaxation times, optimized pulse shaping [52], and further investigation of measurement-induced-state-transition errors. The averaged relaxation times $T_1$ of qubits $Q_1$–$Q_4$ in the current experiments were 45(6), 26(1), 38(1), and 34(1) μs, respectively, at the time of the readout benchmarking. We note that these relaxation times—which we believe to be limited by dielectric loss—are lower than the predicted Purcell-limited values shown in Fig. 5 for the case without intrinsic notch filtering. Thus, we expect that the assignment fidelities demonstrated here could be achieved using large-linewidth filter resonators without intrinsic notch filtering. Nevertheless, the intrinsic notch filtering will prevent Purcell relaxation from becoming a significant loss channel for transmon qubits with state-of-the-art relaxation times.

The results demonstrate that tailoring readout modes to have large externally limited linewidths and dispersive shifts is an effective strategy to increase assignment fidelities, in agreement with Refs. [14] and [17]. Here, the four multiplexed resonators consume a bandwidth of approximately 500 MHz due to the large linewidths. The multiplexing factor could be increased further by replacing the IMPA amplifier with a traveling-wave parametric amplifier (TWPA), where 20 dB gain has been demonstrated over bandwidths exceeding 2 GHz [53,54].

A potential drawback of using large linewidths and dispersive shifts is that it increases the susceptibility of qubits to dephasing due to the noise-photon population in the readout resonator [55–57]. However, we expect some degree of protection from this dephasing channel in this device due to the high readout-resonator frequencies, which are all above 10 GHz. This is because a thermal photon population in the readout resonator is expected to follow Bose-Einstein statistics [58] and thus be exponentially suppressed with increasing frequency at a given temperature. The Hahn-echoed pure dephasing times $T_{2\mathrm{echo}}$ of qubits $Q_1$ to $Q_4$ are 61, 55, 152, and 77 μs. Attributing this dephasing entirely to the noise-photon population $n_r^{\mathrm{noise}}$ of the readout resonators sets upper

bounds of $n_r^{\text{noise}} \leq \{3.1, 3.2, 1.0, 2.4\} \times 10^{-4}$ (for details, see Appendix E). These values correspond to effective temperatures of between 55 and 62 mK as a result of the high readout-resonator frequencies. It may be possible to further suppress the sensitivity to noise-photon induced dephasing by introducing weak anharmonicity to the filter resonators [19].

## IV. CONCLUSIONS

We have shown that by coupling a quarter-wavelength readout resonator and filter resonator together, both capacitively and inductively, it is possible to engineer an intrinsic notch filter that can provide strong auxiliary Purcell filtering to the qubit. An equivalent-circuit model has been introduced and used to find analytic expressions for the notch filtering and for the coupling strength between the readout and filter resonators. We have experimentally verified these formulas in a device with four-to-one readout multiplexing. The notch filter broadens the range of linewidths for the readout mode that are possible without impacting the qubit relaxation time. By designing readout modes with linewidths as large as 42 MHz, we have tuned up a fast multiplexed readout across four qubits using a 56-ns measurement pulse and integration window and benchmarked an average simultaneous assignment fidelity of 99.77%, with the highest qubit assignment fidelity equal to 99.91%. With the simulated readout ring-down time included, the total readout duration for the four qubits ranges from 115 ns to 215 ns, which we expect can be reduced to approximately 100 ns with active ring-down pulse shaping. The compact coupled resonators with intrinsic notch filtering offer strong Purcell protection and are straightforward to design, making this an attractive circuit component for superconducting-qubit readout.

## ACKNOWLEDGMENTS

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## APPENDIX A: DEVICE AND EXPERIMENTAL SETUP

The four-qubit unit cell measured in this work is from the 16-qubit device shown in Fig. 8(a). The qubits are coupled to nearest neighbors (NNs) by short transmission-line sections, resulting in an effective capacitive coupling between nearest neighbors. The average NN coupling strength is 7.0 MHz and the NN static *ZZ* crosstalk ranges between 80 and 200 kHz. The circuit back side where the qubit- and readout-drive wirings make contact with the chip is shown in Fig. 8(b). In Fig. 8(c), we show a close-up of the TSV contacted by a readout pogo pin that routes readout signals to the circuit front side.

The device was cooled to approximately 20 mK inside a Bluefors XLD1000 dilution refrigerator. The experimental

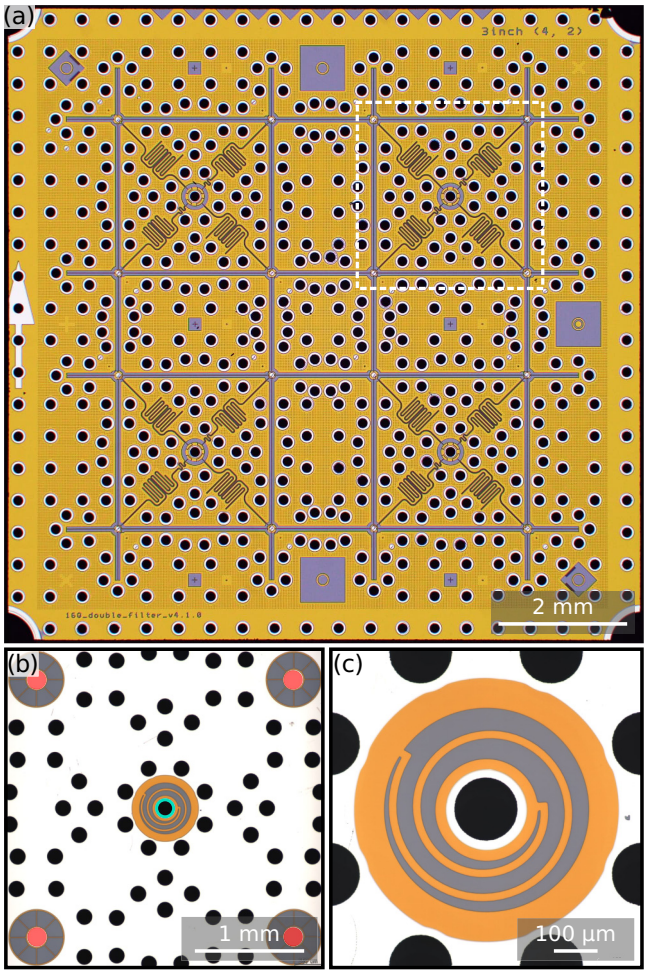



FIG. 8. Images of the 16-qubit device. (a) Photograph of the 16-qubit chip. The four-qubit unit cell measured in this work is highlighted. (b) False-color image of the back side of the four-qubit unit cell, showing the contact pads where pogo pins contact the chip for qubit control (red) and for multiplexed qubit readout (teal). (c) Image of the TSV contacted by the readout pogo pin, showing the TiN spiral shunt inductor.

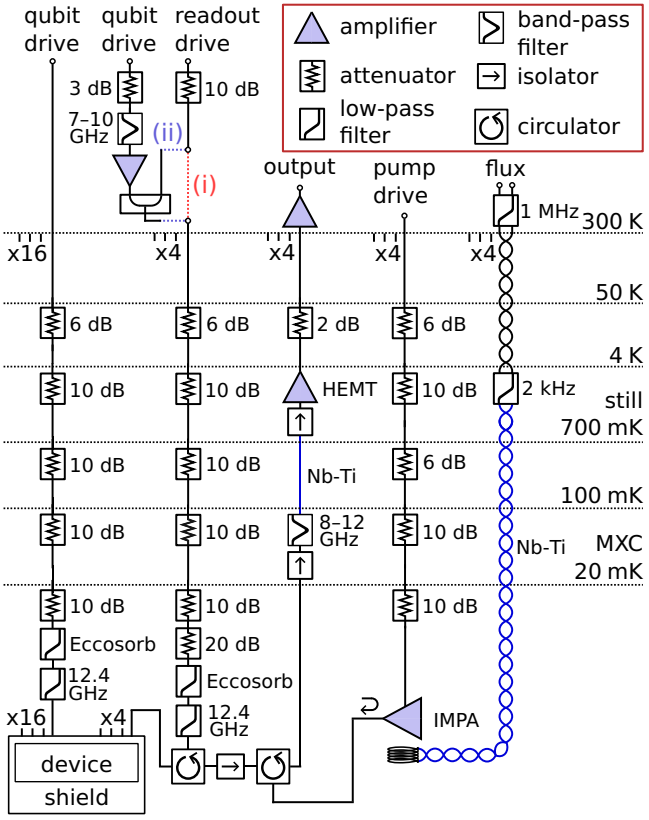



FIG. 9. Experimental setup. The readout drive line is wired in two configurations. In configuration (i), the readout drive is connected directly to the input line. In configuration (ii), it is combined with an amplified qubit drive before connecting to the input line. The IMPA amplifier refers to the impedance-matched parametric amplifier. The term MXC refers to the mixing-chamber stage.

setup is shown in Fig. 9. For the qubit and readout drives, the impedance-matched parametric amplifier pump drive, and the readout-signal acquisition, a QuEL-1 controller (QuEL Inc. [59]) is used. For the experiments characterizing the Purcell filtering, the readout drive is combined with a qubit drive, which is used to generate the Stark-shifting tone. In this configuration, a room-temperature amplifier (Mini-circuits ZVA-183W-S+) is attached to the Stark-shifting qubit drive in order to increase the Stark shift induced in the qubit.

## APPENDIX B: TRANSFER IMPEDANCE BETWEEN TWO MTL-COUPLED λ/4 RESONATORS

In this appendix, we derive an analytic solution for the transfer impedance between two MTL-coupled $\lambda/4$ resonators. We outline the necessary assumptions for achieving a tractable solution and demonstrate that these assumptions are well justified for the coupled CPW lines in this

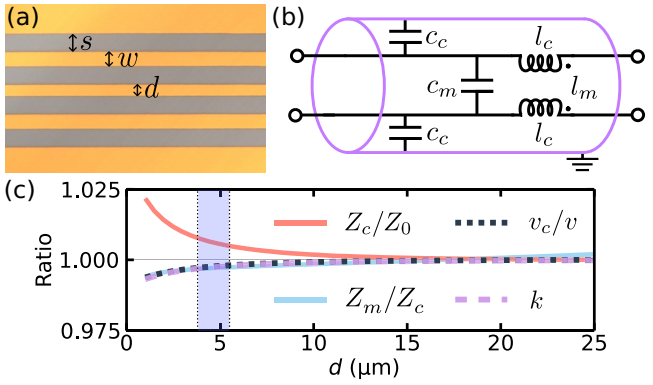



FIG. 10. Analysis of the coupled CPW lines. (a) Image of the coupled CPW lines used to realize the MTL section. (b) Definitions for the per-length parameters of the two-line MTL. (c) Finite-element electrostatic simulation results using COMSOL for the coupled CPW lines, showing the dimensionless quantities of the MTL lines as a function of the width $d$ of the separating ground strip. The values of $d$ for the coupled CPW lines in this work lie inside the blue-highlighted region.

work. We discuss the general solution as well as the specific cases of lumped-element capacitive coupling and MTL coupling in a homogeneous medium.

In Fig. 10(a), we show an image of two coupled CPW lines of the kind used in this work to couple the readout and filter resonators. The key line dimensions are the ground separation $s = 7.5\,\mu\text{m}$, the center-line width $w = 5\,\mu\text{m}$, and the width of the ground strip that separates the two CPWs, $d$, which takes values between $3.8\,\mu\text{m}$ and $5.5\,\mu\text{m}$.

The two coupled CPW lines are treated as an MTL with per-length parameters as defined in Fig. 10(b). In order to find a tractable analytic solution for the transfer impedance $Z_{21}(\omega)$ between two $\lambda/4$ resonators coupled together by this MTL, the following assumptions are applied:

(i) Cyclic symmetric lines. The two lines of the MTL are both taken to have the same capacitance to ground and self-inductance per length, here denoted by $c_c$ and $l_c$, respectively.

(ii) Weakly coupled lines. An excitation applied to one line of the MTL will induce an excitation in the other line. The weak-coupling assumption amounts to neglecting the backaction of this induced excitation onto the first line and is valid when $l_m \ll l_c$ and $c_m \ll c_c$. In a homogeneous medium, this is relaxed to the less restrictive condition $k \equiv (1 - (l_m/l_c)^2)^{1/2} \approx 1$ [60].

(iii) Consonant lines. The two lines of the MTL have characteristic impedance and phase velocity defined as $Z_c \equiv \sqrt{l_c/(c_c + c_m)}$ and $v_c \equiv 1/\sqrt{l_c(c_c + c_m)}$, respectively. Here, the term "consonant line" is defined to mean that these quantities equal the characteristic impedance and phase velocity for the

uncoupled sections of the $\lambda/4$ resonators, such that $Z_c = Z_0$ and $v_c = v$.

In Fig. 10(c), we present COMSOL finite-element simulation results showing that the weak-coupling and consonant-line assumptions are well justified for the coupled CPW lines considered in this work. The simulation results also confirm the relationship $Z_m = Z_c$ for two coupled CPW lines, where $Z_m \equiv \sqrt{l_m/c_m}$. As discussed later in Appendix B2, this relationship is characteristic of lines that are embedded in a homogeneous medium.

In Fig. 11(a), we show the distributed circuit used to model the coupled resonators, with a current source attached to port 1. The transfer impedance between ports 1 and 2 is defined as $Z_{21}(\omega) \equiv V_2(\omega)/I_1(\omega)|_{I_2=0}$. In order to solve for this, the problem is broken down into parts as depicted in Figs. 11(b)–11(d). First, the ratio $V_A(\omega)/I_1(\omega)$ is solved for, where $V_A(\omega)$ is the voltage at port A of the MTL. Using the weak-coupling assumption to neglect backaction and using the consonant-line assumption to set $Z_c = Z_0$ and $v_c = v$, the problem reduces to that of solving for the voltage at a distance $\ell_r^o$ from the open end of a shorted transmission line of total length $\ell_r = \ell_r^o + \ell_c + \ell_r^s$, as depicted in Fig. 11(b). This yields

$$\frac{V_A(\omega)}{I_1(\omega)} = iZ_0\left[\tan\left(\frac{\omega\ell_r}{v}\right)\cos\left(\frac{\omega\ell_r^o}{v}\right) - \sin\left(\frac{\omega\ell_r^o}{v}\right)\right]. \tag{B1}$$

Next, the voltage $V_C(\omega)$ at port C of the MTL line is solved for in terms of the voltage $V_A(\omega)$, as depicted in Fig. 11(c). Here, we make use of the general solution presented in Ref. [60, Eq. (26a)] for a weakly coupled two-line MTL

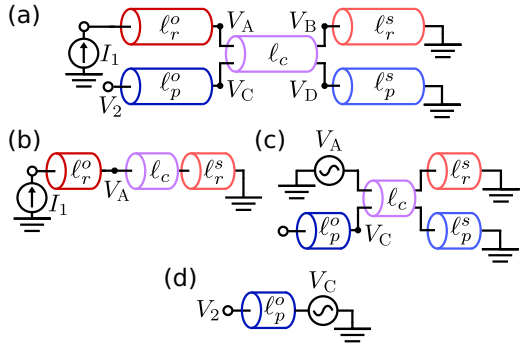



FIG. 11. Solution steps for determining the transfer impedance $Z_{21}(\omega)$. (a) Distributed-circuit model for two MTL-coupled $\lambda/4$ resonators. A current source drives port 1 and the goal is to determine the induced voltage $V_2$ at port 2. The induced voltages at the four ports of the MTL are defined as $V_A$–$V_D$. (b) Circuit used to find the voltage $V_A$ in terms of the current drive $I_1$. (c) Circuit used to find the voltage $V_C$ in terms of the voltage drive $V_A$. (d) Circuit used to find the voltage $V_2$ in terms of the voltage drive $V_C$.

terminated with arbitrary loads. We use the consonant-line assumption to set $Z_c = Z_0$ and $v_c = v$ and after some algebraic manipulations arrive at the following expression:

$$\frac{V_C(\omega)}{V_A(\omega)} = Z_0\omega\ell_c c_m \frac{(A_+ - A_-)\cos\left(\frac{\omega\ell_p^o}{v}\right)}{2\cos\left(\frac{\pi\omega}{2\omega_p}\right)\sin\left[\frac{\omega(\ell_r^s+\ell_c)}{v}\right]}, \tag{B2}$$

where

$$A_+ = \left(1 + Z_m^2/Z_0^2\right)\mathrm{sinc}\left(\frac{\omega\ell_c}{v}\right)\cos\left[\frac{\omega\left(\ell_r^s + \ell_p^s + \ell_c\right)}{v}\right], \tag{B3}$$

$$A_- = \left(1 - Z_m^2/Z_0^2\right)\cos\left[\frac{\omega\left(\ell_r^s - \ell_p^s\right)}{v}\right]. \tag{B4}$$

Here, we restate that $Z_m \equiv \sqrt{l_m/c_m}$. Finally, the voltage $V_2$ is solved for in terms of the voltage $V_C$ as depicted in Fig. 11(d), yielding

$$\frac{V_2(\omega)}{V_C(\omega)} = \sec\left(\frac{\omega\ell_p^o}{v}\right). \tag{B5}$$

The transfer impedance is found by combining these solutions:

$$Z_{21}(\omega) = \left(\frac{V_2}{V_C}\right)\left(\frac{V_C}{V_A}\right)\left(\frac{V_A}{I_1}\right). \tag{B6}$$

Substituting in Eqs. (B1), (B2), and (B5), the overall expression simplifies to yield the final result

$$Z_{21}(\omega) = iZ_0^2\omega\ell_c c_m \frac{A_+ - A_-}{2\cos\left(\frac{\pi\omega}{2\omega_r}\right)\cos\left(\frac{\pi\omega}{2\omega_p}\right)}. \tag{B7}$$

This solution has the required properties of being purely imaginary and reciprocal such that $Z_{12}(\omega) = Z_{21}(\omega)$. This approach to solving for the transfer impedance can also be applied to find tractable solutions in the case in which one of the resonators is mirrored through a line bisecting the MTL section, as well as in the cases in which one (both) of the resonators is (are) replaced with $\lambda/2$ resonators. Furthermore, under the weak-coupling and consonant-line assumptions, the solution generalizes to resonators that are connected by multiple MTL sections. In this case, the transfer impedance is found by summing the contributions coming from each MTL section treated independently, as depicted in Fig. 12. In this way, Eq. (B7) can also be used for modeling the response and predicting notch frequencies for resonators that are coupled by multiple MTL sections.

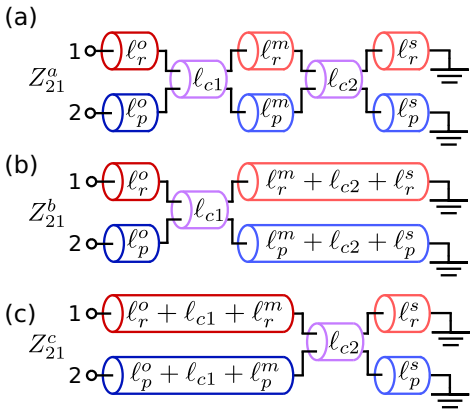



FIG. 12. (a) Pair of $\lambda/4$ resonators coupled by two MTL sections. The transfer impedance is defined as $Z^a_{21}$. (b),(c) Same pair of resonators coupled by only the (b) first and (c) second MTL sections, respectively. Under the weak-coupling and consonant-line assumptions, the transfer impedances satisfy the relation $Z^a_{21} = Z^b_{21} + Z^c_{21}$.

### 1. Capacitive coupling

As stated in the main text, lumped-element capacitive coupling represents a special case of the MTL coupling found by taking the limits $c_m \ell_c \to C_J$ and $l_m, \ell_c \to 0$. In this case, the solution for $Z_{21}(\omega)$ given in Eq. (B7) reduces to

$$Z_{21}(\omega) = -iZ_0^2 \frac{\sin\left(\frac{\omega \ell^s_r}{v}\right)\sin\left(\frac{\omega \ell^s_p}{v}\right)}{\cos\left(\frac{\pi\omega}{2\omega_r}\right)\cos\left(\frac{\pi\omega}{2\omega_p}\right)}\omega C_J. \tag{B8}$$

This expression has its first zeros at frequencies $\pi v/\ell^s_r$ and $\pi v/\ell^s_p$. These terms satisfy $\pi v/\ell^s_r \geq 2\omega_r$ and $\pi v/\ell^s_p \geq 2\omega_p$, leading to the bound on the notch frequency $\omega_n \geq \min(2\omega_r, 2\omega_p)$ that is stated in the main text.

### 2. MTL coupling

For a two-line cyclic symmetric MTL in a homogeneous medium with relative permittivity $\epsilon_r$, the per-length capacitances and inductances are constrained to obey to relations [32]

$$l_c(c_c + c_m) - l_m c_m = \mu_0 \epsilon_0 \epsilon_r, \tag{B9}$$

$$Z_m = Z_c. \tag{B10}$$

Substituting Eq. (B10) into Eq. (B7) and using the consonant-line assumption to set $Z_c = Z_0$, the solution for $Z_{21}(\omega)$ reduces to

$$Z_{21}(\omega) = iZ_0 \frac{\sin\left(\frac{\omega \ell_c}{v}\right)\cos\left(\frac{\pi\omega}{2\omega_n}\right)}{\cos\left(\frac{\pi\omega}{2\omega_r}\right)\cos\left(\frac{\pi\omega}{2\omega_p}\right)}\left(\frac{c_m}{c}\right). \tag{B11}$$

Here, $\omega_n$ is the notch frequency at which $Z_{21}(\omega) = 0$, which we restate takes the form

$$\omega_n = \frac{\pi v}{2\left(\ell^s_r + \ell_c + \ell^s_p\right)}. \tag{B12}$$

## APPENDIX C: EQUIVALENT LUMPED-ELEMENT CIRCUITS AND INTERACTION HAMILTONIANS

In this appendix, we provide a mapping between the distributed and equivalent lumped-element circuits shown in Fig. 13. The mapping is chosen such that the equivalent circuits closely match the electrical response of the distributed circuits around the readout- and filter-resonator frequencies. The calculated transfer impedances for the distributed and equivalent circuits are shown in Fig. 3(c) for the particular set of circuit parameters given in Table III, indicating good agreement. Given that the electrical responses closely match, we infer that the interaction Hamiltonians for both the distributed and equivalent circuits are also closely matched.

We first review that a $\lambda/4$ transmission-line resonator with length $\ell$ has its fundamental resonance at $\omega_{\lambda/4} = \pi v/(2\ell)$ and that at frequencies near this resonance, the electrical response is well approximated by a parallel $LC$ resonator with lumped-element values given by [61]

$$C_{\lambda/4} = \frac{c\ell}{2}, \quad L_{\lambda/4} = \frac{8l\ell}{\pi^2}, \tag{C1}$$

where $c$ and $l$ are the capacitance and inductance per length of the line, respectively, satisfying $c = 1/(Z_0 v)$ and $l = Z_0/v$. Using this result, the lumped-element capacitance and inductance of the readout resonator and filter resonator

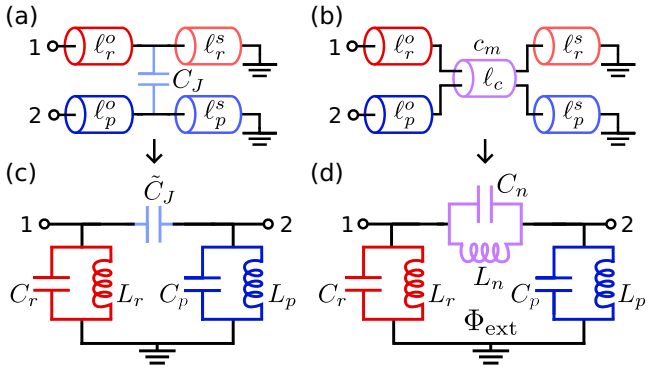



FIG. 13. Distributed circuits and corresponding equivalent lumped-element circuits for the coupled readout and filter resonators. The equivalent-circuit models are used to determine the coupling strength between the $\lambda/4$ modes of the readout resonator and filter resonator, for the cases of (a) [(c)] lumped-element capacitive coupling, and (b) [(d)] MTL coupling.

TABLE III. Readout- and filter-resonator dimensions used to calculate the transfer impedances plotted in Fig. 3(c). The "Cap" ("MTL") parameters are used to calculate the transfer impedance for the distributed circuit with the lumped-element coupling capacitor (MTL coupler). The value of $C_J$ is set to 1.4 fF and the value of $c_m$ is set to 8.5 fF/mm in order to match the coupling strengths. Values of $v = 1.19 \times 10^8$ m/s and $Z_0 = 66\,\Omega$ have been used for the phase velocity and characteristic impedance of the lines. The MTL parameters match the designed parameters for the readout and filter resonators of $Q_1$.

| | $\ell_r^o$ (μm) | $\ell_r^s$ (μm) | $\ell_p^o$ (μm) | $\ell_p^s$ (μm) | $\ell_c$ (μm) | $d$ (μm) |
|---|---|---|---|---|---|---|
| Cap | 1133 | 1776 | 918 | 1818 | ⋯ | ⋯ |
| MTL | 974 | 1617 | 759 | 1659 | 318 | 5.5 |

are mapped from the distributed circuit by the relations

$$C_{r(p)} = \frac{\ell_{r(p)}}{2Z_0 v}, \quad L_{r(p)} = \frac{8Z_0\ell_{r(p)}}{\pi^2 v}. \tag{C2}$$

The characteristic impedances of the lumped-element resonators, defined as $Z_{r(p)} \equiv \sqrt{L_{r(p)}/C_{r(p)}}$, are related to the transmission-line characteristic impedance by the formula

$$Z_{r(p)} = \frac{4Z_0}{\pi}. \tag{C3}$$

### 1. Capacitive coupling

In Fig. 13(c), we depict the equivalent circuit for the capacitively coupled $\lambda/4$ resonators. Equating the transfer impedance $Z_{21}^{\rm ec}(\omega)$ for the equivalent circuit with the transfer impedance for the distributed circuit in Fig. 13(a) [given by Eq. (B8)] yields the following expression for the ratio between the equivalent-circuit capacitance $\tilde{C}_J$ and the distributed-circuit capacitance $C_J$:

$$\frac{\tilde{C}_J}{C_J} = \frac{\pi^2}{16}\frac{\left(\frac{\omega_r}{\omega} - \frac{\omega}{\omega_r}\right)\left(\frac{\omega_p}{\omega} - \frac{\omega}{\omega_p}\right)}{\cos\left(\frac{\omega\pi}{2\omega_r}\right)\cos\left(\frac{\omega\pi}{2\omega_p}\right)}\sin\left(\frac{\ell_r^s\omega}{v}\right)\sin\left(\frac{\ell_p^s\omega}{v}\right), \tag{C4}$$

where the weak-coupling assumption $\tilde{C}_J \ll C_r, C_p$ is used. This ratio is frequency dependent. However, the frequency dependence is weak around the $\lambda/4$-mode frequencies $\omega_r$ and $\omega_p$. In order to determine a frequency-independent value for $\tilde{C}_J$ that is accurate around these modes, the expression is separated into readout-resonator- and filter-resonator-dependent factors and these are then evaluated with $\omega$ set to the corresponding resonance frequencies:

$$\frac{\tilde{C}_J}{C_J} \approx \left(\frac{\pi}{4}\right)^2 A_r A_p, \tag{C5}$$

$$A_r = \left[\frac{\left(\frac{\omega_r}{\omega} - \frac{\omega}{\omega_r}\right)}{\cos\left(\frac{\omega\pi}{2\omega_r}\right)}\sin\left(\frac{\ell_r^s\omega}{v}\right)\right]\Bigg|_{\omega\to\omega_r}, \tag{C6}$$

$$A_p = \left[\frac{\left(\frac{\omega_p}{\omega} - \frac{\omega}{\omega_p}\right)}{\cos\left(\frac{\omega\pi}{2\omega_p}\right)}\sin\left(\frac{\ell_p^s\omega}{v}\right)\right]\Bigg|_{\omega\to\omega_p}. \tag{C7}$$

Evaluating the factors yields

$$\tilde{C}_J \approx C_J \sin\left(\frac{\ell_r^s\omega_r}{v}\right)\sin\left(\frac{\ell_p^s\omega_p}{v}\right). \tag{C8}$$

According to this approximation, the equivalent-circuit capacitance vanishes as the distributed-circuit capacitor approaches the shorted end of either $\lambda/4$ resonator and takes its maximum value when the distributed-circuit capacitor is connected to the open end of both resonators, in which case $\tilde{C}_J = C_J$.

The equivalent lumped-element circuit in Fig. 13(c) is solved by the branch-flux method [62,63] to yield the following interaction Hamiltonian between the readout-resonator and dedicated-filter modes under the RWA:

$$\hat{\mathcal{H}}_{\rm int} = \hbar J^{\rm cap}\left(\hat{a}_r^\dagger\hat{a}_p + \hat{a}_r\hat{a}_p^\dagger\right), \tag{C9}$$

$$J^{\rm cap} = \frac{1}{2}\sqrt{Z_r Z_p}\,\omega_r\omega_p\tilde{C}_J. \tag{C10}$$

The solution is valid assuming weak coupling between the modes such that $\tilde{C}_J \ll C_r, C_p$. Substituting in Eq. (C3) for the characteristic impedances $Z_{r(p)}$ and Eq. (C8) for the capacitance $\tilde{C}_J$ yields

$$J^{\rm cap} = \frac{2}{\pi}Z_0\omega_r\omega_p C_J \sin\left(\frac{\omega_r\ell_r^s}{v}\right)\sin\left(\frac{\omega_p\ell_p^s}{v}\right). \tag{C11}$$

This expression gives the coupling strength $J^{\rm cap}$ purely in terms of the distributed-circuit parameters.

### 2. MTL coupling

In Fig. 13(d), we depict the equivalent circuit for the MTL-coupled $\lambda/4$ resonators. The form of the circuit is motivated as the coupling capacitance $C_n$ and coupling inductance $L_n$ destructively interfere, resulting in a notch frequency at which transmission through the coupled resonators is suppressed. To determine a mapping for the coupling elements $C_n$ and $L_n$, we match the transfer impedance $Z_{21}^{\rm ec}$ for the equivalent circuit with the transfer impedance $Z_{21}$ for the distributed circuit in Fig. 13(b) [given by Eq. (B11)] around the notch frequency $\omega_n$. We

equate both the zeroth- and first-order derivatives,

$$Z_{21}^{\mathrm{ec}}(\omega_n) = Z_{21}(\omega_n) = 0, \tag{C12}$$

$$\left.\frac{dZ_{21}^{\mathrm{ec}}(\omega)}{d\omega}\right|_{\omega=\omega_n} = \left.\frac{dZ_{21}(\omega)}{d\omega}\right|_{\omega=\omega_n}. \tag{C13}$$

The first equation gives the condition $\sqrt{L_n C_n} = 1/\omega_n$ and the second equation gives the condition

$$Z_n \equiv \sqrt{\frac{L_n}{C_n}} = Z_0 \frac{64}{\pi^3} \frac{\cos\left(\frac{\pi\omega_n}{2\omega_r}\right)\cos\left(\frac{\pi\omega_n}{2\omega_p}\right)}{\left(\frac{\omega_r}{\omega_n} - \frac{\omega_n}{\omega_r}\right)\left(\frac{\omega_p}{\omega_n} - \frac{\omega_n}{\omega_p}\right)} \frac{(c/c_m)}{\sin\left(\frac{\omega_n \ell_c}{v}\right)}. \tag{C14}$$

The effective Hamiltonian of the MTL-coupled resonators can now be found because the Hamiltonian of the equivalent lumped-element circuit can be constructed following a standard procedure [62,63]. The external flux $\Phi_{\mathrm{ext}}$ through the inductive loop is set to zero, as the distributed circuit from which the equivalent circuit is constructed has no such inductive loop. Focusing on the interaction part of the Hamiltonian and making the RWA yields

$$\hat{\mathcal{H}}_{\mathrm{int}} = \hbar J \left(\hat{a}_r^\dagger \hat{a}_p + \hat{a}_r \hat{a}_p^\dagger\right), \tag{C15}$$

$$J = \frac{\sqrt{Z_r Z_p}}{2Z_n} \sqrt{\omega_r \omega_p} \left(\frac{\sqrt{\omega_r \omega_p}}{\omega_n} - \frac{\omega_n}{\sqrt{\omega_r \omega_p}}\right). \tag{C16}$$

The solution is valid assuming weak coupling between the modes such that $C_n \ll C_r, C_p$ and $L_n \gg L_r, L_p$. The solutions to $Z_{r(p)}$ and $Z_n$ given in Eqs. (C3) and (C14), respectively, are substituted into this expression. In addition, we make the change of variables $\omega_r = \overline{\omega}_{rp} + \Delta_{rp}/2$ and $\omega_p = \overline{\omega}_{rp} - \Delta_{rp}/2$, where $\overline{\omega}_{rp}$ is the average of the resonator and filter frequencies and $\Delta_{rp}$ is the detuning. Expanding in orders of the term $\Delta_{rp}/\omega_n$ yields

$$J = \overline{\omega}_{rp} \frac{\pi^2}{32} \frac{\left(\frac{\overline{\omega}_{rp}}{\omega_n} - \frac{\omega_n}{\overline{\omega}_{rp}}\right)^3}{\cos^2\left(\frac{\pi\omega_n}{2\overline{\omega}_{rp}}\right)} \left(\frac{c_m}{c}\right) \sin\left(\frac{\omega_n \ell_c}{v}\right) + \mathcal{O}\left[\left(\frac{\Delta_{rp}}{\omega_n}\right)^2\right]. \tag{C17}$$

For readout applications, it is typically desirable to make the detuning $\Delta_{rp}$ as small as possible and so the higher-order terms can be neglected. Dropping these terms then results in Eq. (5), which gives the exchange coupling strength between two MTL-coupled $\lambda/4$ resonators in terms of the resonator frequencies, the notch frequency, and the length and coupling capacitance of the MTL coupler.

## APPENDIX D: CIRCUIT MODELS FOR THE PURCELL-LIMITED RELAXATION TIME

In this appendix, the circuit models used to predict the Purcell-limited qubit relaxation time $T_1^{\mathrm{pl}}$ are described. The models are then used to derive an expression for the Purcell-filtering enhancement provided by the intrinsic notch filter.

In Fig. 14, we show the general circuit used to model the energy relaxation of the qubit due to its coupling to the readout line. The qubit, with shunt capacitance $C_q$, is coupled via a capacitor with capacitance $C_{qr}$ to a lossless two-port network. The network is coupled at port 2 to a readout line with characteristic impedance $Z_{\mathrm{line}}$ via a capacitor with capacitance $C_{\mathrm{ext}}$. The characteristic relaxation time of the qubit imposed by its coupling to the readout line can be modeled by the following equation [34]:

$$T_1^{\mathrm{pl}}(\omega_q) = \frac{C_q}{\mathrm{Re}[Y_{\mathrm{in}}(\omega_q)]}. \tag{D1}$$

Defining $Z_{11}$, $Z_{22}$, and $Z_{21}$ as the impedance matrix elements for the lossless two-port network, the real part of the input admittance $Y_{\mathrm{in}}$ for the circuit in Fig. 14 is given by

$$\mathrm{Re}[Y_{\mathrm{in}}(\omega)] = \frac{Z_{\mathrm{line}}\,|Z_{21}|^2}{\left|\left(Z_{11} + Z_{qr}\right)\left(Z_{22} + Z_{\mathrm{ext}} + Z_{\mathrm{line}}\right)\right|^2}, \tag{D2}$$

$$Z_{qr} = \frac{-i}{\omega C_{qr}}, \quad Z_{\mathrm{ext}} = \frac{-i}{\omega C_{\mathrm{ext}}}. \tag{D3}$$

This expression is valid when $Z_{21} \ll Z_{11}, Z_{22}$. The transfer impedances $Z_{21}$ for the two-port network given the specific circuits shown in Figs. 13(c) and 13(d) are defined as $Z_{21}^{\mathrm{cap}}$ and $Z_{21}^{\mathrm{MTL}}$, respectively. Likewise, the Purcell-limited relaxation times of the qubit given these specific circuits are defined as $T_1^{\mathrm{cap}}$ and $T_1^{\mathrm{MTL}}$.

The predicted Purcell-limited relaxation times plotted in Fig. 5(b) are generated using $T_1^{\mathrm{cap}}$ and $T_1^{\mathrm{MTL}}$. The circuit parameters are constrained to match the measured frequencies, coupling strengths, and linewidths for the qubit and the readout and filter resonators, in a similar manner to Ref. [31]. For completeness, the shunting impedance to ground given by Eq. (E7) in Appendix E is included in the

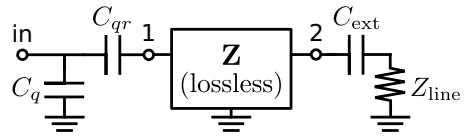



FIG. 14. Circuit model for determining the Purcell-limited relaxation time of a qubit. A qubit with shunt capacitance $C_q$ is capacitively coupled to port 1 of a lossless two-port network with impedance matrix **Z**. Port 2 of the network is capacitively coupled to a load with resistance $Z_{\mathrm{line}}$.

model. The effect of the shunt impedance is the following adjustment to the values of $Z_{\rm ext}$ and $Z_{\rm line}$:

$$Z_{\rm ext} \to Z_{\rm ext} + \frac{Z_{\rm shunt}}{1+|Z_{\rm shunt}/Z_{\rm line}|^2}, \tag{D4}$$

$$Z_{\rm line} \to \frac{Z_{\rm line}}{1+|Z_{\rm line}/Z_{\rm shunt}|^2}. \tag{D5}$$

Including the shunt impedance in the model results in a modest increase in the Purcell-limited relaxation times at the qubit frequencies of around 5–10%.

For weak coupling between the readout resonator and dedicator filter and at frequencies away from their resonances, $Z_{11}$ and $Z_{22}$ are, to a good approximation, independent of the coupling-circuit implementation such that $Z_{11}^{\rm cap} = Z_{11}^{\rm MTL}$ and $Z_{22}^{\rm cap} = Z_{22}^{\rm MTL}$. As a result, the ratio of the relaxation times takes the form

$$\frac{T_1^{\rm MTL}(\omega_q)}{T_1^{\rm cap}(\omega_q)} = \left|\frac{Z_{21}^{\rm cap}(\omega_q)}{Z_{21}^{\rm MTL}(\omega_q)}\right|^2. \tag{D6}$$

Given that the readout resonator and dedicated filter are weakly coupled, this reduces to

$$\frac{T_1^{\rm MTL}(\omega_q)}{T_1^{\rm cap}(\omega_q)} = \left[\frac{\omega_q Z_n \tilde{C}_J}{\left(\frac{\omega_n}{\omega_q}-\frac{\omega_q}{\omega_n}\right)}\right]^2. \tag{D7}$$

Using Eqs. (C10) and (C16) from Appendix C, the parameters $\tilde{C}_J$ and $Z_n$ can be expressed in terms of the coupling strengths $J^{\rm cap}$ and $J^{\rm MTL}$, respectively, yielding

$$\tilde{C}_J = \frac{\pi J^{\rm cap}}{2Z_0\overline{\omega}_{rp}^2} + \mathcal{O}\left[\left(\frac{\Delta_{rp}}{\overline{\omega}_{rp}}\right)^2\right], \tag{D8}$$

$$Z_n = \frac{2}{\pi}Z_0\left(\frac{\overline{\omega}_{rp}}{\omega_n}-\frac{\omega_n}{\overline{\omega}_{rp}}\right)\frac{\overline{\omega}_{rp}}{J^{\rm MTL}} + \mathcal{O}\left[\left(\frac{\Delta_{rp}}{\omega_n}\right)^2\right]. \tag{D9}$$

Dropping the higher-order terms and substituting these expressions into Eq. (D7) expresses the ratio of the relaxation times in terms of frequencies and coupling strengths. The coupling strengths $J^{\rm cap}$ and $J^{\rm MTL}$ are then equated so as to ensure a fair comparison between the $T_1$ limits, yielding

$$\frac{T_1^{\rm MTL}(\omega_q)}{T_1^{\rm cap}(\omega_q)} = \frac{1}{4}\frac{\omega_q^2}{\Delta_{qn}^2}\left(1-\frac{\omega_n^2}{\overline{\omega}_{rp}^2}\right)^2\left(1+\mathcal{O}\left[\left(\frac{\Delta_{qn}}{\omega_n}\right)^2\right]\right), \tag{D10}$$

where $\Delta_{qn} = \omega_q - \omega_n$ is the detuning of the qubit from the notch. Dropping the higher-order terms and defining $\xi \equiv T_1^{\rm MTL}(\omega_q)/T_1^{\rm cap}(\omega_q)$ results in Eq. (6).

We now find an approximate expression for the Purcell-filtering enhancement factor $\xi_{\rm filter} \equiv T_1^{\rm cap}/T_1^{\rm ref}$ provided by the filter resonator. This factor compares $T_1^{\rm cap}$ to the Purcell-limited relaxation time $T_1^{\rm ref}$ expected when the readout resonator is directly coupled with linewidth $\kappa_{\rm ro}$ to the readout line. In analogy with Eq. (D6), we find the following expression for this ratio in terms of the circuit-model parameters:

$$\frac{T_1^{\rm cap}(\omega_q)}{T_1^{\rm ref}(\omega_q)} = \left|\frac{Z_r(\omega_q)}{Z_{21}^{\rm cap}(\omega_q)}\right|^2\left(\frac{C_{\rm ext,ro}}{C_{\rm ext}}\right)^2, \tag{D11}$$

where $C_{\rm ext,ro}$ is the capacitance that couples the readout resonator directly to the readout line resulting in a linewidth $\kappa_{\rm ro}$ and

$$Z_r(\omega_q) = \frac{4iZ_0}{\pi\left(\frac{\omega_r}{\omega_q}-\frac{\omega_q}{\omega_r}\right)}. \tag{D12}$$

Here, we assume weak coupling to the line such that $|Z_{\rm ext}(\omega_q)| \gg |Z_r(\omega_q)|, Z_{\rm line}$ and $|Z_{\rm ext,ro}(\omega_q)| \gg |Z_p(\omega_q)|, Z_{\rm line}$. To proceed, we express the ratio of the external capacitances in terms of the linewidths $\kappa_{\rm ro}$ and $\kappa_p$, yielding

$$\frac{T_1^{\rm cap}(\omega_q)}{T_1^{\rm ref}(\omega_q)} = \left|\frac{Z_r(\omega_q)}{Z_{21}^{\rm cap}(\omega_q)}\right|^2\left(\frac{\kappa_{\rm ro}}{\kappa_p}\right). \tag{D13}$$

To arrive at this result, we again assume weak coupling to the line and further assume that the detuning between the readout and filter resonator is small, such that $\Delta_{rp} \ll \overline{\omega}_{rp}$. Finally, we assume that the readout and filter resonator are weakly coupled and make use of Eq. (D8) to express $\tilde{C}_J$ in terms of the coupling strength $J$, arriving at the expression

$$\xi_{\rm filter} = \frac{1}{4}\left(\frac{\overline{\omega}_{rp}}{J}\right)^2\left(1-\frac{\overline{\omega}_{rp}^2}{\omega_q^2}\right)^2\left(\frac{\kappa_{\rm ro}}{\kappa_p}\right). \tag{D14}$$

The total Purcell-filtering enhancement factor $\xi_{\rm total}$ is then given by

$$\xi_{\rm total} = \xi \times \xi_{\rm filter}. \tag{D15}$$

For the device in this work, we predict the total Purcell-filtering enhancement factors to be $\xi_{\rm total} \approx \{12\,200, 235, 35, 15\} \times 10^3$ for qubits $Q_1$–$Q_4$.

## APPENDIX E: SEMICLASSICAL MODEL FOR THE READOUT SYSTEM DYNAMICS

This appendix describes the semiclassical model for the multiplexed readout and filter resonators. The model is utilized to predict both the steady-state and time-dependent responses of the readout system to an external drive, and to determine the normal modes of the system.

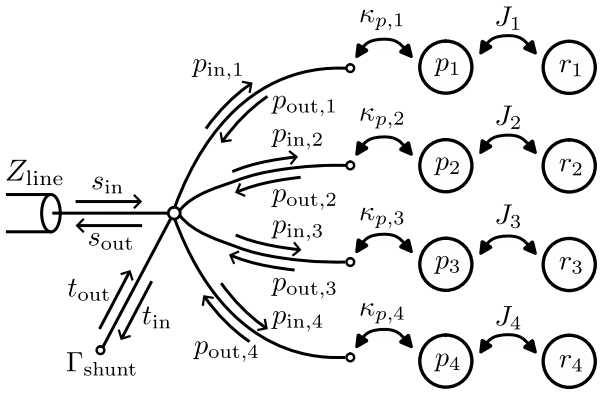



FIG. 15. Input-output network for the multiplexed system of four readout resonators with field amplitudes $r_j$ and dedicated filter resonators with field amplitudes $p_j$ ($j = 1, \ldots, 4$). The multiplexed resonators couple to the end of a transmission line with characteristic impedance $Z_{\mathrm{line}}$. The model is generalized to include a shunt impedance to ground.

In Fig. 15, we show the input-output network used to model the multiplexed readout system. A traveling coherent field $s_{\mathrm{in}}$ propagating along a transmission line with characteristic impedance $Z_{\mathrm{line}}$ is incident on the system of coupled readout and filter resonators. The traveling fields are treated classically when considering their behavior at the shunt and at the node at which the fields scatter. They are treated using input-output theory [64] when considering their interaction with the resonators. We denote the coherent field amplitudes of filter resonator $j$ and readout resonator $j$ as $p_j$ and $r_j$, respectively. The term $\kappa_{p,j}$ is the externally limited linewidth of filter mode $j$ and the term $J_j$ is the coupling strength between filter resonator $j$ and readout resonator $j$. The model is linear and qubits are treated as only inducing a state-dependent shift to the readout-resonator frequencies.

The reflection coefficient of the incident field is defined as

$$\Gamma_{\mathrm{incident}} \equiv \frac{s_{\mathrm{out}}}{s_{\mathrm{in}}}. \tag{E1}$$

The incident field is treated as separating into distinct fields that each interact with a single filter mode or with the shunt. The incoming and outgoing fields at the node at which the separation occurs obey the relations

$$s_{\mathrm{out}} - s_{\mathrm{in}} = t_{\mathrm{out}} - t_{\mathrm{in}} + \sum_{j=1}^{4} \left(p_{\mathrm{out},j} - p_{\mathrm{in},j}\right), \tag{E2}$$

$$s_{\mathrm{out}} + s_{\mathrm{in}} = t_{\mathrm{out}} + t_{\mathrm{in}} = p_{\mathrm{out},j} + p_{\mathrm{in},j}. \tag{E3}$$

These conditions result from Kirchhoff's laws. The fields incident at the shunt and the filter resonators are reflected with reflection coefficients defined as

$$\Gamma_{\mathrm{shunt}} \equiv \frac{t_{\mathrm{out}}}{t_{\mathrm{in}}}, \quad \Gamma_{p,j} \equiv \frac{p_{\mathrm{out},j}}{p_{\mathrm{in},j}}. \tag{E4}$$

Combining Eqs. (E1)–(E4) yields the following solution for the reflection coefficient of an incident readout signal:

$$\frac{1-\Gamma_{\mathrm{incident}}}{1+\Gamma_{\mathrm{incident}}} = \frac{1-\Gamma_{\mathrm{shunt}}}{1+\Gamma_{\mathrm{shunt}}} + \sum_{j=1}^{4}\left(\frac{1-\Gamma_{p,j}}{1+\Gamma_{p,j}}\right). \tag{E5}$$

We note that this expression is equivalent to stating that the total load impedance terminating the transmission line is the parallel sum of the shunt impedance and the readout-circuit impedance.

The reflection coefficient at the shunt is given by

$$\Gamma_{\mathrm{shunt}} = \frac{Z_{\mathrm{shunt}} - Z_{\mathrm{line}}}{Z_{\mathrm{shunt}} + Z_{\mathrm{line}}}. \tag{E6}$$

Here, the shunt impedance $Z_{\mathrm{shunt}}$ takes the form of a parallel $LC$ resonator

$$\frac{1}{Z_{\mathrm{shunt}}} = -\frac{i}{\omega L_{\mathrm{shunt}}} + i\omega C_{\mathrm{shunt}}. \tag{E7}$$

The term $C_{\mathrm{shunt}}$ is the parasitic shunt capacitance of the TSV structure that routes signals from the pogo pin on the circuit back side to the filter resonators and $L_{\mathrm{shunt}}$ is due to the back-side spiral inductance that connects the TSV structure to the ground plane [Figs. 4(e) and 8(c)]. Shunt values of $C_{\mathrm{shunt}} = 230\,\mathrm{fF}$ and $L_{\mathrm{shunt}} = 1.01\,\mathrm{nH}$ are determined by COMSOL simulations and an analytic formula for the inductance of a spiral inductor [65]. At the frequency $1/(2\pi\sqrt{L_{\mathrm{shunt}}C_{\mathrm{shunt}}}) = 10.44\,\mathrm{GHz}$, the shunt impedance goes to infinity and thus the reflection coefficient $\Gamma_{\mathrm{shunt}}$ approaches 1. As a result, around the readout frequencies, the effect of the large shunt capacitance is screened out by the shunt inductance.

Using input-output theory [64], the incident and reflected fields at the filter resonators are related to the coherent field amplitudes of the filter resonators by the expression

$$p_{\mathrm{out},j} = p_{\mathrm{in},j} - \sqrt{\kappa_{p,j}}\, p_j. \tag{E8}$$

The equations of motion for the coupled filter-mode and readout-resonator-mode system are given by

$$\frac{dp_j}{dt} = -i\Delta_{pd,j}p_j - \frac{\kappa_{p,j}+\gamma_{p,j}}{2}p_j - iJ_j r_j + \sqrt{\kappa_{p,j}}p_{\mathrm{in},j}, \tag{E9}$$

$$\frac{dr_j}{dt} = -i\Delta_{rd,j}^{g(e)}r_j - \frac{\gamma_{r,j}}{2}r_j - iJ_j p_j. \tag{E10}$$

Here, $\Delta_{pd,j} = \omega_{p,j} - \omega_d$ is the detuning of filter mode $j$ from the drive, $\Delta_{rd,j}^{g(e)} = \omega_{r,j}^{g(e)} - \omega_d$ is the qubit-state-dependent detuning of readout mode $j$ from the drive, and

$\omega^e_{r,j} = \omega^g_{r,j} + 2\chi_j$ is the readout-mode frequency given that the qubit is in the $|e\rangle$ state. The terms $\gamma_{r,j}$ and $\gamma_{p,j}$ are the internally limited linewidths of readout resonator $j$ and filter resonator $j$, respectively.

### 1. Steady-state results

Solving the equations of motion in the steady state and combining with Eq. (E8) yields the following solution for the qubit-state-dependent reflection coefficients at the dedicated filters:

$$\Gamma^{g(e)}_{p,j} = 1 - \frac{4i\kappa_{p,j}\left(\Delta^{g(e)}_{rd,j} - i\gamma_{r,j}/2\right)}{\left(2i\Delta_{pd,j} + \kappa_{p,j} + \gamma_{p,j}\right)\left(2i\Delta^{g(e)}_{rd,j} + \gamma_{r,j}\right) + 4J_j^2}. \tag{E11}$$

By combining this result with Eq. (E6) for the reflection coefficient at the shunt, the steady-state reflection at the device given by Eq. (E5) is fully determined from the readout and shunt parameters. An important detail is that input-output theory uses the clockwise sign convention for phasor rotation, whereas Eq. (E7) for the shunt impedance uses the counterclockwise convention from electrical engineering. When fitting to the measured data, we choose the electrical-engineering convention and so all values of $i$ in Eq. (E11) are replaced with $-i$. The measured reflection response $\Gamma_{\text{meas}}$ is fit to the model function

$$\arg\left(\Gamma_{\text{meas}}\right) = \arg\left(\Gamma_{\text{incident}}\right) + \theta_0 - \omega\tau. \tag{E12}$$

Here, $\theta_0$ is a constant phase offset and $\tau$ accounts for the electrical length of the experimental setup. The reflection spectrum $\Gamma_{\text{meas}}$ is measured twice over a frequency range that has covered all of the readout-resonator and filter-resonator modes; first with all qubits prepared in the $|g\rangle$ state and then with all qubits prepared in the $|e\rangle$ state. The two data sets are then fitted simultaneously to extract the parameters $\omega_{r,j}$, $\omega_{p,j}$, $\kappa_{p,j}$, $J_j$, and $\chi_j$ for all the filter resonators and readout resonators. The parameters $\gamma_{p,j}$ and $\gamma_{r,j}$ are set to zero in the model function, since the external coupling of the filter resonators dominated over the internal loss. The bare readout-resonator and filter-resonator parameters extracted from the fit are presented in Table IV.

The bounds on the noise-photon populations $n^{\text{noise}}_{r,j}$ of readout resonators given in the main text are determined from the fitted parameters using the relation [19]

$$\Gamma^{\text{noise}}_{\phi,j} = \frac{n^{\text{noise}}_{r,j}}{2}\int_{-\infty}^{\infty}\left|\Gamma^{e}_{\text{incident}}(\omega) - \Gamma^{g}_{\text{incident}}(\omega)\right|^2\frac{d\omega}{2\pi}, \tag{E13}$$

where $\Gamma^{\text{noise}}_{\phi,j}$ is the noise-photon-induced dephasing rate in qubit $j$ and $\Gamma^{g(e)}_{\text{incident}}(\omega)$ is the reflection coefficient at the device when qubit $j$ is in the $|g\rangle$ and $|e\rangle$ states, respectively.

TABLE IV. Bare readout parameters. Here, $g$ is the coupling strength between the qubit and the readout resonator and $n_{\text{crit}}$ is the critical photon number of the readout resonator.

| | $\omega^g_r/2\pi$ (MHz) | $\omega_p/2\pi$ (MHz) | $J/2\pi$ (MHz) | $\kappa_p/2\pi$ (MHz) | $\chi/2\pi$ (MHz) | $g/2\pi$ (MHz) | $n_{\text{crit}}$ |
|---|---|---|---|---|---|---|---|
| $Q_1$ | 10 250 | 10 232 | 36.1 | 97.6 | −9.4 | 420 | 7.0 |
| $Q_2$ | 10 386 | 10 407 | 39.4 | 81.4 | −9.9 | 423 | 6.7 |
| $Q_3$ | 10 540 | 10 566 | 30.9 | 66.7 | −10.5 | 280 | 7.1 |
| $Q_4$ | 10 666 | 10 710 | 26.2 | 93.5 | −8.3 | 275 | 9.4 |

### 2. Time-dependent results

In order to solve the time dependence of the coherent field amplitudes, it is convenient to use a matrix representation. The driven system of coupled readout and filter resonators can be expressed by the matrix equation

$$\frac{d}{dt}\begin{pmatrix}\mathbf{p}\\ \mathbf{r}\end{pmatrix} = -i\begin{pmatrix}\mathbf{P} & \mathbf{J}\\ \mathbf{J} & \mathbf{R}^{g(e)}\end{pmatrix}\begin{pmatrix}\mathbf{p}\\ \mathbf{r}\end{pmatrix} + \begin{pmatrix}\mathbf{d}\\ \mathbf{0}\end{pmatrix}s_{\text{in}}. \tag{E14}$$

Here, $\mathbf{P}$ is a $4\times 4$ matrix for the filter-resonators, $\mathbf{J}$ is a $4\times 4$ matrix for the couplings between the filter-resonator and readout-resonator modes, and $\mathbf{R}^{g(e)}$ is a qubit-state-dependent $4\times 4$ matrix for the readout resonators. The terms $\mathbf{p}$ and $\mathbf{r}$ are the field amplitudes for the filter resonators and readout resonators, respectively, given by

$$\mathbf{p} = \begin{pmatrix}p_1\\ p_2\\ p_3\\ p_4\end{pmatrix}, \quad \mathbf{r} = \begin{pmatrix}r_1\\ r_2\\ r_3\\ r_4\end{pmatrix}, \tag{E15}$$

and $\mathbf{d}$ gives the coupling coefficients of the driving field $s_{\text{in}}$ to the filter resonators:

$$\mathbf{d} = \left(\frac{1+\Gamma_{\text{shunt}}}{2}\right)\begin{pmatrix}\sqrt{\kappa_{p,1}}\\ \sqrt{\kappa_{p,2}}\\ \sqrt{\kappa_{p,3}}\\ \sqrt{\kappa_{p,4}}\end{pmatrix}. \tag{E16}$$

The matrix $\mathbf{P}$ has elements given by

$$\mathbf{P}_{ij} = \left(\Delta_{pd,i} - i\frac{\gamma_{p,i}}{2}\right)\delta_{ij} - i\frac{\sqrt{\kappa_{p,i}\kappa_{p,j}}}{4}\left(1+\Gamma_{\text{shunt}}\right), \tag{E17}$$

where $\delta_{ij}$ is the Kronecker delta function. The second term introduces off-diagonal elements to the matrix $\mathbf{P}$, which represent coupling between the different filter modes. These off-diagonal elements arise due to the fact that the filter resonators all couple to the readout line via a shared node. The matrix $\mathbf{J}$ is diagonal, with elements given by

$$\mathbf{J}_{ij} = J_i\delta_{ij}. \tag{E18}$$

This matrix couples filter-resonator amplitude $p_j$ to readout-resonator amplitude $r_j$ with coupling strength

$J_j$. Finally, the matrix $\mathbf{R}^{g(e)}$ is diagonal, with elements given by

$$\mathbf{R}_{ij}^{g(e)} = \left(\Delta_{rd,i}^{g(e)} - i\frac{\gamma_{r,i}}{2}\right)\delta_{ij}. \tag{E19}$$

This matrix contains all of the qubit-state dependence and has 16 different permutations depending on the four-qubit state. The system of coupled first-order differential equations in Eq. (E14) can be solved numerically to find the time evolution of the coherent field amplitudes of the readout and filter resonators under a drive. The time-dependent output field $s_{\text{out}}(t)$ into the transmission line is then determined from the filter-resonator amplitudes through Eqs. (E2), (E3) and (E8), yielding

$$s_{\text{out}}(t) = \Gamma_{\text{shunt}} s_{\text{in}}(t) - \left(\frac{1+\Gamma_{\text{shunt}}}{2}\right)\sum_{j=1}^{4}\sqrt{\kappa_{p,j}}\, p_j(t). \tag{E20}$$

The output-field separation $S(t) \equiv |s_{\text{out}}^e(t) - s_{\text{out}}^g(t)|$ for a readout drive targeting qubit $j$ is found by solving this expression. The predicted output-field separation for a readout drive targeting qubit $Q_2$ is plotted in Fig. 7(a).

### 3. Normal modes

In the absence of an external drive, the eigenvalues $\tilde{\lambda}_k$ ($k = 1, \ldots, 8$) of Eq. (E14) give the normal modes of the coupled readout and filter resonators. Here, we set $\Gamma_{\text{shunt}} = 1$, as this is a good approximation around the readout- and filter-resonator frequencies and it simplifies finding the eigenvalues. The frequencies and effective linewidths of the normal modes are then given by

$$\tilde{\omega}_k = \text{Re}\left(\tilde{\lambda}_k\right), \quad \tilde{\kappa}_k = -2\text{Im}\left(\tilde{\lambda}_k\right). \tag{E21}$$

The eight normal modes are separated into pairs, each corresponding to the hybridized modes of a particular readout-resonator and filter-resonator pair. This picture is a slight simplification of the true nature of these modes, since in reality the coupling between the filter resonators means that the normal modes hybridize across the whole network of resonators. However, by looking at the qubit-state dependence, it is clear that each normal mode can be associated with a particular readout-resonator and filter-resonator pair. The dispersive shifts induced by a target qubit are found by solving the normal modes for the $|g\rangle$ and $|e\rangle$ states of the target qubit with all other qubits in the ground state and using the standard definition $\tilde{\chi}_r \equiv (\tilde{\omega}_r^e - \tilde{\omega}_r^g)/2$. The dispersive shift induced in spectator normal modes (mediated via the coupling of the filter resonators) is in all cases less than 1% of the shift in the target normal mode.

TABLE V. Normal modes of the composite readout-resonator and filter-resonator system. The terms $\tilde{\chi}_{r(p)}$ are the dispersive shifts of the normal modes defined as $\tilde{\omega}_{r(p)}^e \equiv \tilde{\omega}_{r(p)}^g + 2\tilde{\chi}_{r(p)}$.

| | $\tilde{\omega}_r^g/2\pi$ (MHz) | $\tilde{\omega}_p^g/2\pi$ (MHz) | $\tilde{\kappa}_r^g/2\pi$ (MHz) | $\tilde{\kappa}_r^e/2\pi$ (MHz) | $\tilde{\kappa}_p^g/2\pi$ (MHz) | $\tilde{\chi}_r/2\pi$ (MHz) | $\tilde{\chi}_p/2\pi$ (MHz) |
|---|---|---|---|---|---|---|---|
| $Q_1$ | 10 221 | 10 284 | 42 | 30 | 42 | −5.9 | −3.5 |
| $Q_2$ | 10 360 | 10 438 | 34 | 25 | 63 | −7.8 | −2.3 |
| $Q_3$ | 10 520 | 10 582 | 24 | 14 | 55 | −8.4 | −2.3 |
| $Q_4$ | 10 652 | 10 701 | 19 | 11 | 59 | −7.2 | −1.1 |

The normal modes are summarized in Table V. Here, $\tilde{\omega}_r^{g(e)}$ denotes the readout-resonator-like modes and $\tilde{\omega}_p^{g(e)}$ the filter-resonator-like modes. By definition, the readout-resonator-like modes have smaller linewidths and larger dispersive shifts than the corresponding filter-resonator-like modes. For the readout characterization, measurement is performed by driving the readout-resonator-like modes, as this results in superior SNR. As a result, we refer to these modes as the readout modes and $\tilde{\omega}_r^g$, $\tilde{\kappa}_r^g$, $\tilde{\chi}_r$ in Table V correspond to $\omega_{\text{ro}}^g$, $\kappa_{\text{ro}}^g$, $\chi_{\text{ro}}$ in Table I.

## APPENDIX F: AC-STARK-SHIFT SPECTROSCOPY MEASUREMENT

In order to calibrate the drive power incident on the device, we follow the protocol in Ref. [17]. We first apply a continuous drive tone to the readout resonator of a particular qubit, in order to induce an ac Stark shift. We subsequently apply a $\pi$ pulse to the qubit followed by qubit measurement. The experiment is then repeated while sweeping the frequency $\omega_d$ of the $\pi$ pulse. The value of $\omega_d$ that results in the maximum qubit population exchange from the $|g\rangle$ to the $|e\rangle$ state is associated with the ac-Stark-shifted qubit frequency. The steady-state average photon number $n_{r,j}^g$ in readout resonator $j$ is then inferred from the ac Stark shift $\Delta_{\text{ac}}$ through the relation $\Delta_{\text{ac}} = 2\chi_j n_{r,j}^g$ [41]. This photon number is related to the readout-resonator coherent field amplitude $r_j^g$ by the expression $n_{r,j}^g = |r_j^g|^2$. The equations of motion given in Eqs. (E9) and (E10) are solved in the steady state to find the input field on the filter resonator, $p_{\text{in},j}$, as a function of $r_g$. Finally, the incident field on the device is determined from the incident field on the filter resonator using the relation

$$\frac{s_{\text{in}}}{p_{\text{in},j}} = \frac{1+\Gamma_{p,j}}{1+\Gamma_{\text{incident}}}. \tag{F1}$$

Here, $\Gamma_{\text{incident}}$ and $\Gamma_{p,j}$ are the reflection coefficients defined in Appendix E, which are determined from the readout and shunt parameters. The squared norm of the incident field $|s_{\text{in}}|^2$ corresponds to the photon flux $\dot{n}$ on the device and thus the power incident on the device is given by $P = \hbar\omega_d|s_{\text{in}}|^2$.

From the calibrated power incident on the device, we determine the attenuation between the generator and the device. We make the assumption that this attenuation is the same over the range of frequencies that are driven in the subsequent section. Since the attenuation of the coaxial cables of the input lines increases with frequency, this assumption will lead to lower-bound values for the Purcell-limited relaxation time.

Next, the drive amplitude $\Omega$ incident on the qubit is determined as a function of frequency. Here, $\Omega$ is defined in terms of the drive Hamiltonian as $\hat{\mathcal{H}}_{\rm drive} = \hbar\Omega\,(a_q^\dagger + a_q)\cos(\omega_d t)$, where $a_q^\dagger$ and $a_q$ are the creation and annihilation operators for the qubit.

As described in the main text, $\Omega$ is determined by two methods following Ref. [16]: (i) by measuring the Rabi frequencies of the qubit when driving at the $|g\rangle$-$|e\rangle$ and $|e\rangle$-$|f\rangle$ transition frequencies and (ii) by measuring the induced ac Stark shift $\Delta_{\rm ac}$ in the qubit when detuned from these transitions. The drive is applied through the readout line. In Figs. 16(a) and 16(b), we show Rabi oscillations in $Q_1$. The $|g\rangle$-$|e\rangle$ and $|e\rangle$-$|f\rangle$ Rabi oscillation frequencies are equal to $\Omega$ and $\sqrt{2}\,\Omega$, respectively.

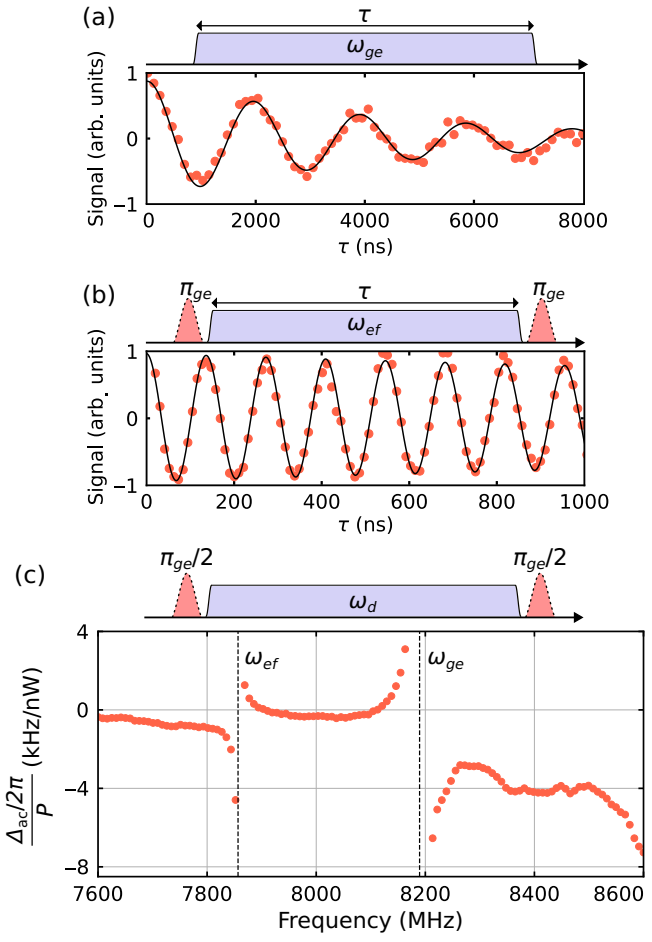



FIG. 16. (a),(b) Rabi oscillations in qubit $Q_1$ driven at the (a) $|g\rangle$-$|e\rangle$ and (b) $|e\rangle$-$|f\rangle$ transition frequencies, respectively. (c) Pulse sequence for measuring the ac Stark shift and the results for qubit $Q_2$. The ratio between the Stark shift and the incident power $P$ is given as a function of the drive frequency.

In Fig. 16(c), we show the measured ac Stark shift $\Delta_{\rm ac}$ normalized by the incident power for qubit $Q_2$ as a function of the drive frequency. The drive amplitude $\Omega$ is determined from the measured ac Stark shift using the perturbative formula in Ref. [16]. The results are sensitive to the calibrated value of the qubit frequency $\omega_q$. We determined this more precisely by applying a Stark-shifting drive to the qubit over a range of powers $P$ and performing a linear fit of the Stark-shifted qubit frequency $\omega_{q,\rm ac}$ to the formula $\omega_{q,\rm ac} = \omega_q + kP$, where $k$ is the slope of the linear fit. From the fit, we extract the value of $\omega_q$ and an associated uncertainty.

In qubits $Q_1$, $Q_2$, and $Q_3$, we observe a frequency range between the $|g\rangle$-$|e\rangle$ and $|e\rangle$-$|f\rangle$ transition frequencies where the drive induces a small negative ac Stark shift in the qubit—contrary to the positive shift that is expected across this frequency range. This anomalous shift may result from spurious harmonics of the generator producing the Stark-shifting drive, since the sensitivity to the main drive tone is strongly suppressed around the notch frequency. We exclude the data in this range when determining the Purcell-limited relaxation time.

From the measured drive amplitude and the incident power, the Purcell-limited relaxation time of the qubit is given by [16]

$$T_1^{\rm pl}(\omega_d) = \frac{4P}{\Omega^2\hbar\omega_d}. \tag{F2}$$

## APPENDIX G: FURTHER ANALYSIS OF THE FAST MULTIPLEXED QUBIT READOUT

Here, further details of the fast multiplexed readout measurement are provided. First, the calculation of the separation error and coherence limited errors is discussed. The separation error is determined from the SNR of the measurements. Here, the SNR is defined as

$$\mathrm{SNR} \equiv \frac{|\mu_g - \mu_e|}{(\sigma_g + \sigma_e)/2}, \tag{G1}$$

where $\mu_{g(e)}$ and $\sigma_{g(e)}$ are the mean and standard deviation, respectively, of the Gaussian fits to the $|g(e)\rangle$ state histograms shown in Fig. 7(c). The measured SNR values for the four qubits are provided in Table VI. The ratio $\sigma_e/\sigma_g$ is also given, showing that the noise for the $|g\rangle$ and $|e\rangle$ states was almost the same. The separation error due to overlap of the projected Gaussian distributions is estimated from

TABLE VI. Further information regarding the readout. Here, $\sigma_e/\sigma_g$ expresses the ratio of the readout-signal standard deviations for the $|g\rangle$ and $|e\rangle$ states. The terms $P_0(e_2 \mid g_1)$ and $P_\pi(g_2 \mid g_1)$ are the error contributions to the assignment fidelity when the qubit is prepared in the $|g\rangle$ and $|e\rangle$ states, respectively. The term $P_e$ is the excited-state population of the qubits prior to state preparation and $\omega_d$ is the drive frequency that is used to perform readout.

| | SNR | $\sigma_e/\sigma_g$ | $P_0(e_2 \mid g_1)$ (%) | $P_\pi(g_2 \mid g_1)$ (%) | $P_e$ (%) | $\omega_d/2\pi$ (MHz) |
|---|---|---|---|---|---|---|
| $Q_1$ | 6.3 | 1.03 | 0.22 | 0.49 | 1.7 | 10 224 |
| $Q_2$ | 8.4 | 0.99 | 0.03 | 0.16 | 2.3 | 10 357 |
| $Q_3$ | 6.0 | 1.03 | 0.13 | 0.44 | 1.4 | 10 515 |
| $Q_4$ | 6.7 | 0.99 | 0.08 | 0.33 | 1.0 | 10 646 |

the SNR as [17,66]

$$\varepsilon_{\text{sep}} = \frac{1}{2}\left[1 - \text{erf}\left(\frac{\text{SNR}}{\sqrt{8}}\right)\right]. \tag{G2}$$

The coherence limit to the assignment fidelity is estimated as

$$\varepsilon_{\text{cl}} = \frac{\tau_{\text{meas}}}{4T_1}, \tag{G3}$$

where the factor of 4 in the denominator arises because qubit relaxation only occurs when the qubit is prepared in the $|e\rangle$ state and only relaxation events occurring within the first half of the integration window will result in significant state misassignment.

The coherence limit to the QND fidelity is estimated as

$$\varepsilon_{\text{cl}}^{Q} = \frac{\tau_{\text{buffer}}}{2T_1} + \frac{\tau_{\text{meas}}}{2T_1}, \tag{G4}$$

where $\tau_{\text{buffer}}$ is the length of the buffer period between the first and second measurements of the QND measurement-pulse sequence.

In Fig. 17, we show the IQ plots for the simultaneous measurement on the four qubits. These measurements have been postselected to ensure that the qubits are prepared in the $|g\rangle$ state prior to the $\pi$-pulse sequence. Table VI gives the fraction of discarded measurements, represented by the excited-state population $P_e$. In Fig. 17(a), all of the approximately $8 \times 10^4$ postselected shots are plotted. We fit the measured $|g\rangle$- and $|e\rangle$-state data to bivariate normal distributions. From the fits, we have determined the confidence ellipses. In Fig. 17(b), we show the $1\sigma$ (68.27%), $2\sigma$ (95.45%), $3\sigma$ (99.73%) and $4\sigma$ (99.994%) ellipses. We stress that this $\sigma$ is not referring to a standard deviation. Rather, the interpretation is that 68.27% of the single-shot measurements are expected to lie inside the $1\sigma$ confidence ellipse, 95.45% inside the $2\sigma$ ellipse, etc. Shots that have been misassigned are highlighted. These are divided into two categories. Shots that have been misassigned and that lie inside (outside) the $4\sigma$ ellipses for

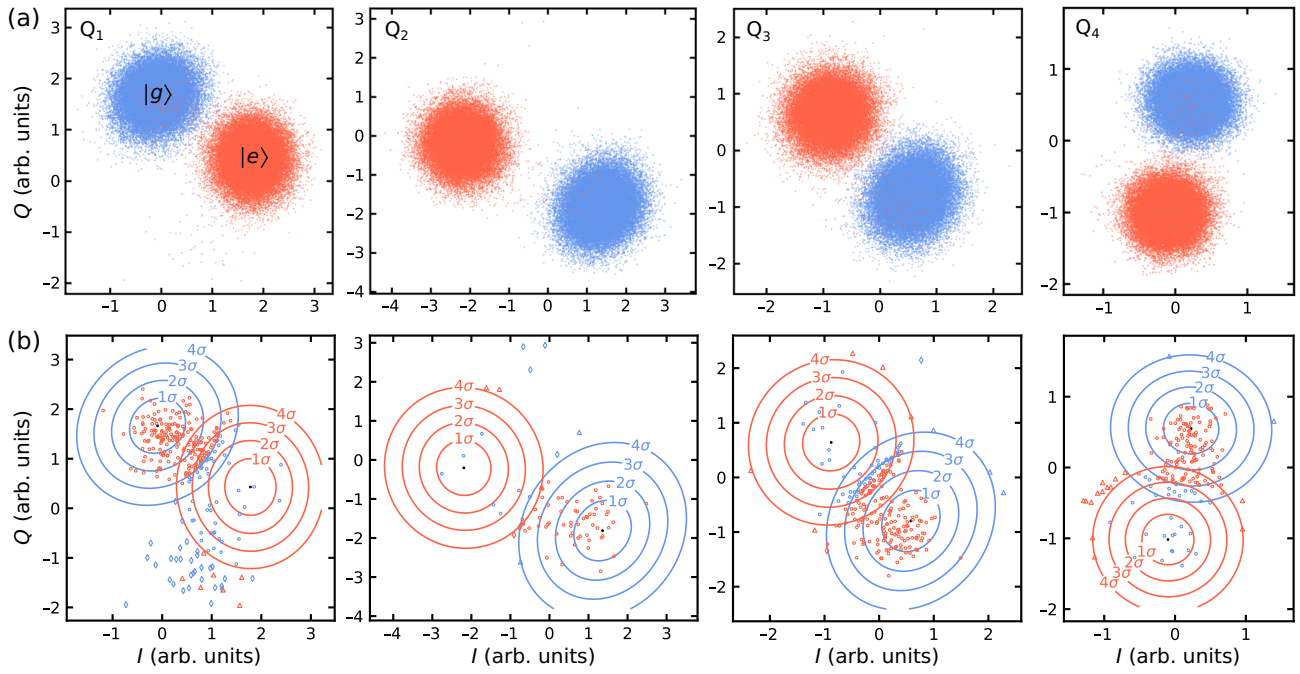



FIG. 17. Postselected IQ signals from the assignment-fidelity pulse sequence. (a) Time-integrated IQ signals in which the qubit is prepared in the $|g\rangle$ state (blue) and in the $|e\rangle$ state (red). (b) The $1\sigma$–$4\sigma$ confidence ellipses of the fitted bivariate normal distributions. Misassigned shots that lie inside either (outside both) of the $4\sigma$ ellipses are shown as circles (diamonds). Correctly assigned shots that lie outside the $4\sigma$ ellipse of the assigned state are shown as triangles.

both the $|g\rangle$ and $|e\rangle$ states are shown as circles (diamonds). We attribute misassigned shots outside the $4\sigma$ ellipses to measurement-induced excitation to leakage states, as we would not expect these events if the qubit were restricted to the $|g\rangle$-$|e\rangle$ subspace. In addition, correctly assigned shots that lie outside the $4\sigma$ ellipse of the assigned state are shown as triangles. We expect approximately five of these events across $8 \times 10^4$ measurements. When many of these events cluster together, we again attribute it to measurement-induced leakage.

For $Q_1$, there is a clear signature of leakage from the $|g\rangle$ state that has made a significant contribution to the assignment error. There is also a signature of leakage from the $|e\rangle$ state that has not led to assignment errors. In the remaining qubits, $>4\sigma$ measurements with signatures of leakage affect less than 0.015% of the total measurements. However, we emphasize that the absence of high-$\sigma$ events in the IQ plane does not preclude all leakage events. For instance, it is possible for the output field given the qubit is in a leakage state to overlap with the $|g\rangle$- or $|e\rangle$-state output fields. Further characterization such as pseudosyndrome detection [29] could be performed in order to detect such "hidden" leakage events.


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